SPE-229918-MS

Successful Design and Execution of Dual Leg Selective and Matrix Acidizing Using Open Hole Inflatable Packer for the First Time in a Horizontal Carbonate Well From one of the Iranian Southwest oil Fields: A Case Study

M. Roshani, A. Zareiforoush, and A. Salimi, Pezhvak Energy Engineering & Services Company, Tehran, Iran; M. Behnood, Behtash Sana'at Arta Kish, Tehran, Iran; P. Molavi, Pezhvak Energy Engineering & Services Company, Tehran, Iran

 DOI 10.2118/229918-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper presents a successful stimulation case in a dual-leg horizontal well in the A formation carbonate reservoir, Target field, Iran. The objective is to demonstrate the effectiveness of combining open-hole inflatable packer technology with coiled tubing-conveyed acidizing to selectively stimulate each lateral separately for the first time in Iran. The study evaluates the impact of selective and matrix acidizing on injectivity, productivity, and operational efficiency.

A two steps acid stimulation was executed. In the first step, selective acidizing was performed by isolating each leg using an open-hole inflatable packer deployed on coiled tubing to perform acid wash. In the second step, after post-CT acid wash cleanup, matrix acidizing using plain acid and VDA, enhanced reservoir connectivity through full lateral coverage. Operational steps were supported by integrated simulations using Acid Stimulation Software and Coiled Tubing Hydraulic and Stress Analysis Simulator ensuring accurate fluid placement, retrievable packer setting depths, and pressure-force limits. Post-job well performance was evaluated via injectivity tests and results, multi-choke well testing and production monitoring to assess the effectiveness of the stimulation.

The stimulation delivered significant performance improvements. After selective acidizing, the injectivity index (II) improved from 0.8 to 2.0 STB/(D·psi), and after matrix acidizing, it reached 4.0 STB/(D·psi)—a fivefold overall increase. The surface productivity index (PI) increased from 1.75 to 8.7 STB/(D·psi), while wellhead pressure increased from 170 psi to 870 psi in the same choke size. Oil production improved from 1,800 BOPD to 2,900 BOPD. The use of the inflatable packer enabled precise acid placement in each leg without mechanical failures, packer lock-up, or coiled tubing issues. Diverter efficiency effectively targeting low-permeability sections evaluating by using an advanced analysis methodology. Compared to matrix acidizing alone, the selective approach removed near-wellbore damage more efficiently, while matrix acidizing ensured deeper penetration across the carbonate formation. The integration of both methods resulted in optimal stimulation coverage with improved flow distribution. The operation was successfully executed under a rig-based setup, demonstrating that complex open-hole

dual-lateral wells can be effectively stimulated using a combination of mechanical isolation and chemical diversion without compromising operational safety or treatment effectiveness.

This case introduces a novel, dual-lateral stimulation methodology in carbonates, combining mechanical isolation via open-hole inflatable packers deployed through coiled tubing with chemical diversion using VDA pumped through the kick-off string. The workflow, backed by real-time operational simulations and post-job validation, sets a new benchmark for complex horizontal well interventions. The approach enhances stimulation precision, reduces operational risks, and delivers significant cost savings, offering a scalable solution for similar carbonate reservoirs worldwide.

Keywords: Two Legs Horizontal Well, Open Hole Inflatable Packer, Well Stimulation and Intervention, Well Testing, Coiled Tubing

Introduction

Stimulation of carbonate reservoirs, particularly in horizontal and dual-lateral wells, remains one of the most technically demanding operations in well intervention and production enhancement. The heterogeneous nature of carbonate formations, coupled with the complexity of open-hole multilateral geometries, often leads to inefficient acid placement, poor reservoir contact, and suboptimal well performance. Traditional matrix acidizing approaches, while effective in simpler wellbores, struggle to provide uniform treatment in such challenging environments.

To overcome these limitations, engineers have explored various chemical diversion techniques, such as viscoelastic surfactant (VES) and self-diverting acids (SDA), as well as mechanical isolation tools, including straddle packers and coiled tubing-conveyed sleeves. However, the field application of such methods in complex open-hole dual-leg wells has remained limited, primarily due to operational risks, mechanical failures, and uncertain treatment coverage.

Acid stimulation in complex horizontal and multilateral carbonate wells has evolved significantly with the integration of mechanical and chemical diversion techniques. Nasr-El-Din et al. introduced a coiled tubing-conveyed orienting tool for indexing into laterals, coupled with self-diverting acid to ensure deeper penetration without mechanical isolation, though it had limitations in lateral reach [1]. Glasbergen and Buijse optimized diversion by using a fluid placement simulator, demonstrating that conventional acid diversion strategies often fall short in heterogeneous formations [2]. Jin (2007) emphasized the importance of pump rate optimization along with chemical diversion to improve acid coverage across long laterals [3]. These findings informed the current approach where open-hole inflatable packers enhance selective acidizing in dual-leg horizontals.

The importance of real-time monitoring tools like Distributed Temperature Sensing (DTS) was highlighted by Glasbergen et al, where uneven acid placement necessitated operational adjustments[4]. Similarly, Sarma et al. demonstrated the use of real-time diversion control with SDA and DPRAS in multilayered carbonate fields, eliminating the need for coiled tubing and mechanical isolation in more than 70 wells [5].

The use of viscoelastic surfactant (VES) systems has become central to modern diversion strategies. Studies such as Nasr-El-Din et al. and Al-Otaibi et al. detailed the effect of VES concentration, permeability contrast, and salinity on diversion efficiency[6], [7]. Chang et al. extended these techniques to high-permeability deepwater sandstones, demonstrating VES's ability to temporarily block dominant flow paths and improve zonal sweep[8]. Czupski et al. (2020) further validated VES-based selective acidizing in dolomitic reservoirs, reducing water cut while targeting oil-bearing zones [9].

Advanced acidizing models and simulators support better acid placement and treatment design. Czupski et al. (2024) applied a simulation-driven HCl/VDA sequence that doubled gas production while minimizing water production in a carbonate gas field [10]. Similarly, Alameedy et al. used Acid Stimulation Software to match modeled skin reduction with field results, highlighting the importance of model-assisted design [11].

Jeffry et al. used nodal analysis and historical data to select optimal acid formulations, a method aligning with the simulation-based workflow of the current study [12].

Matrix acidizing in carbonates must balance dissolution efficiency with formation preservation. The 2021 MDPI paper emphasized the role of acid additives and the integration of mechanical isolation in improving stimulation precision[13]. Al Mahasneh et al. tailored acid compositions based on reservoir heterogeneity and used corrosion inhibitors and environmentally friendly agents to enhance recovery in Jordan's Azraq field [14].

High-temperature carbonate reservoirs require special considerations. Salah Al-Harthy reviewed acid systems stable under high temperatures and proposed combining mechanical isolation with high-performance acids—concepts directly applied in the inflatable packer-based stimulation method used in this case study [15]. Moreover, wormhole propagation modeling, as presented in Buijse et al. provides insights into dissolution behavior and helped guide acid sequencing for effective near-wellbore cleanup[16].

The 1994 Virginia Hills study (Bennion) underlined how drilling and completion damage in horizontal wells can be mitigated through mechanical isolation and targeted stimulation—validating the use of inflatable packers to protect against formation impairment [17].

Finally, Ragi Poyyara et al. (2014) reviewed the full spectrum of diversion strategies across carbonate and sandstone formations, reinforcing that matching diversion type with reservoir properties is key to successful stimulation [18].

This study represents the first successful field application of a dual-phase stimulation strategy in a dual-leg horizontal carbonate well in southwest Iran, integrating selective and matrix acidizing with open-hole inflatable packer technology deployed via coiled tubing. By combining mechanical isolation during the selective acidizing phase with chemical diversion in the matrix stage, this workflow enabled precise acid placement in each lateral and ensured deep reservoir penetration across the entire interval. Beyond demonstrating the operational feasibility of inflatable packers in open-hole geometries, the study highlights how simulation-driven design, customized acid systems (including plain HCl and viscoelastic diverting acid), and real-time operational adjustments can deliver substantial gains in injectivity, productivity, and overall well performance. The results provide a replicable and scalable blueprint for treating complex horizontal carbonate wells, setting a new benchmark for future stimulation campaigns in similar reservoirs worldwide.

Field and Well Description

The X-Y Field is a complex carbonate oil field located in the Abadan Plain of southwestern Iran, adjacent to the prolific SA and Y fields within the broader West Karun basin. Structurally, the field comprises two gentle culminations—X and Y—spanning approximately 352 square kilometers. While the X culmination is primarily developed in the deep B Formation, the Y culmination targets the shallower A carbonate reservoir.

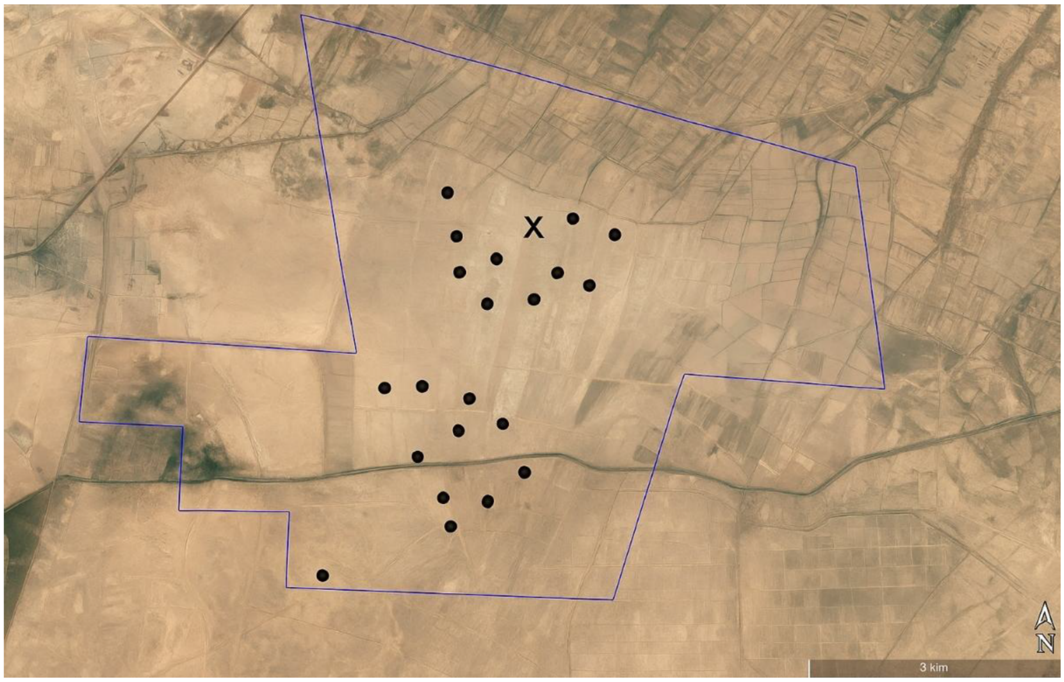


Figure 1—Geographical location of the X-Y field in the Abadan Plain, southwest Iran, highlighting its proximity to SA and Y fields.

The B Formation is a high-pressure, high-temperature (HPHT) reservoir encountered at depths ranging from 4300 to 4600 meters. It is composed of clean, micro-fractured limestones with reservoir temperatures exceeding 280°C and pressures above 10,000 psi. Production wells in B are typically completed in cased-hole configuration with 7-inch casing and deliver initial oil rates between 4,000 and 6,000 barrels per day of light oil (~39–40° API). Standard acid wash and matrix acidizing techniques have proven effective in maintaining productivity in this interval.

In contrast, the A Formation—targeted in the Y culmination—is a shallower, low-permeability carbonate reservoir encountered between approximately 2970- and 4120-meters measured depth. It consists of dense, mud-supported limestones with moderate microporosity and minimal natural fracturing. Stratigraphically, the formation is subdivided into A-Top, A-IIA, and A-IIIB units, with the A-Main (Il-IIIB) considered the primary production zone due to its relatively higher effective porosity and hydrocarbon saturation. The oil produced from this formation is heavier, with an API gravity around 22°, and is associated with high-salinity formation water (110,000–130,000 ppm) and CO_2 concentrations of approximately 1.36 mol%.

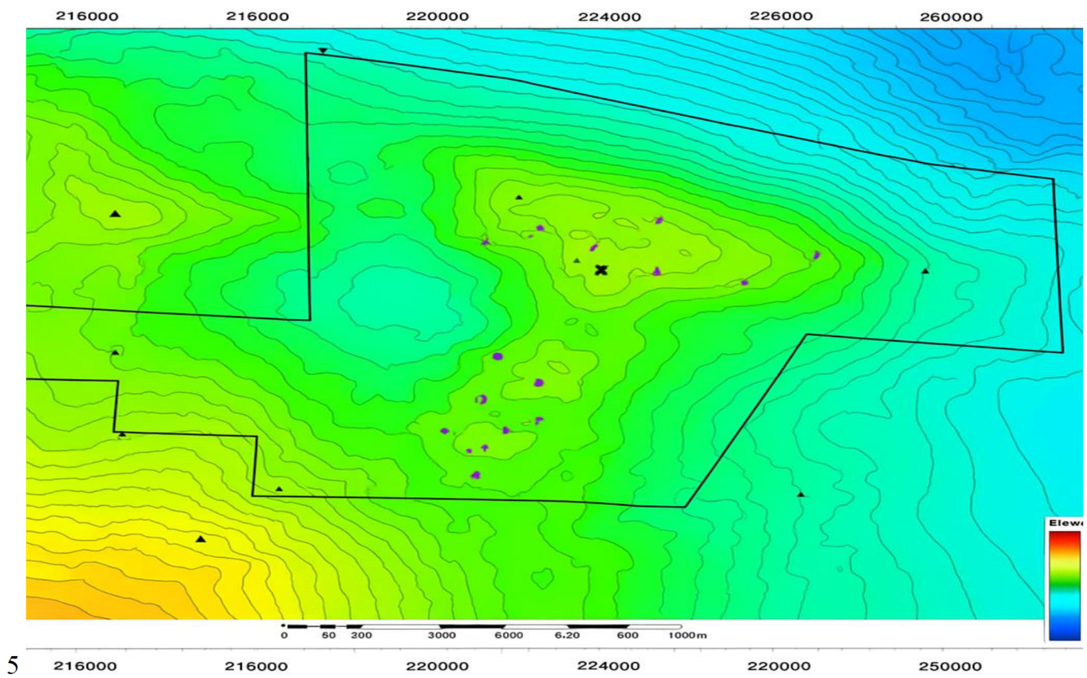


Figure 2—Structural depth map of the X-Y field showing A reservoir geometry and structural culminations.

Well X-1 is a dual-leg open-hole horizontal producer located on the northwestern flank of the Y culmination, strategically positioned near offset wells X-03, X-04, and X-05. The well was geosteered based on updated A Main structural mapping to maximize reservoir contact and enhance drainage efficiency across heterogeneous intervals. X-1 was drilled with two laterals: the main leg, which extends from 3073 to 4056.6 meters measured depth with a 6-1/8-inch open-hole section, and the kick-off leg, which spans from 3225 to 3705 meters measured depth with a 5-7/8-inch open-hole section. A 7-inch liner was set above the kickoff point at 3073 meters. Due to its dual-leg geometry, open-hole completion, and proximity to structurally diverse compartments, X-1 was selected as the pilot well for testing a hybrid stimulation program combining selective and matrix acidizing techniques.

The geological characteristics of the A reservoir directly shaped the stimulation design. The low matrix permeability (1–5 mD) and heterogeneous porosity distribution necessitated the use of a viscoelastic diverting acid (VDA) system to improve diversion and ensure acid coverage beyond high-permeability streaks. Meanwhile, the open-hole completion configuration created the need for mechanical isolation, making coiled tubing-deployed inflatable packers the optimal solution for selectively treating each lateral independently. These packers enabled precise zonal isolation during the selective acidizing phase, while the matrix acidizing stage leveraged alternating HCl and VDA sequences to propagate wormholes deeper into the formation, enhancing long-term reservoir connectivity.

Given the low-permeability nature of the A Formation and limited effectiveness of conventional bullhead acidizing in such heterogeneous carbonates, X-1 was selected for a pilot selective acid stimulation campaign. The treatment employed coiled tubing-conveyed inflatable packers to enable mechanical isolation and stage-wise acid placement in each leg independently. This was followed by matrix acidizing using a combination of hydrochloric acid (HCl 15%) and viscoelastic diverting acid (VDA 15%) to improve stimulation depth and distribution. The results of this case are intended to serve as a benchmark for future completions in the A reservoir and provide a scalable stimulation strategy for similar tight carbonate environments

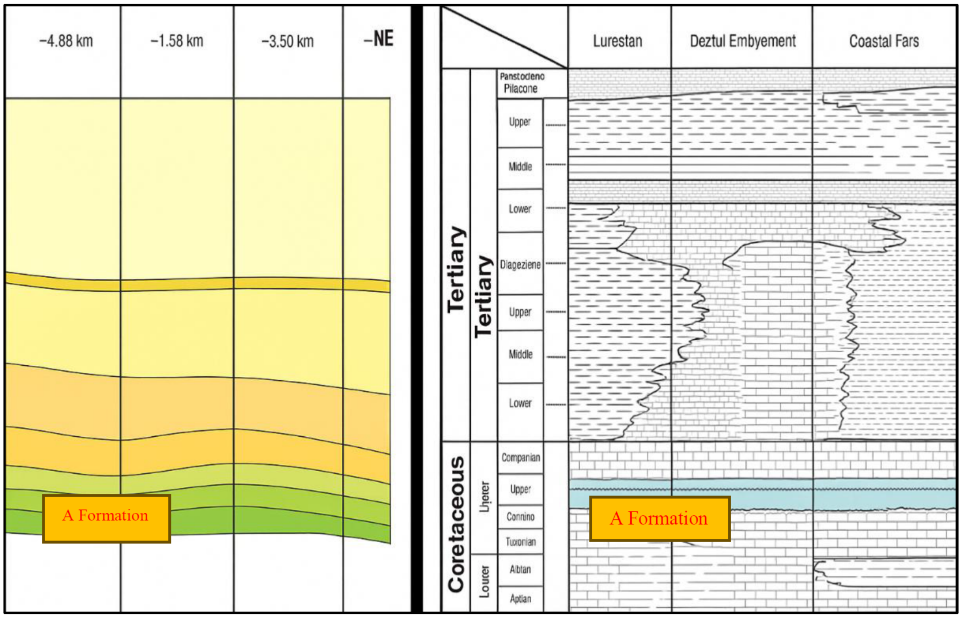


Figure 3—Regional stratigraphic correlation of the X-Y field illustrating the A and B formations and key reservoir intervals.

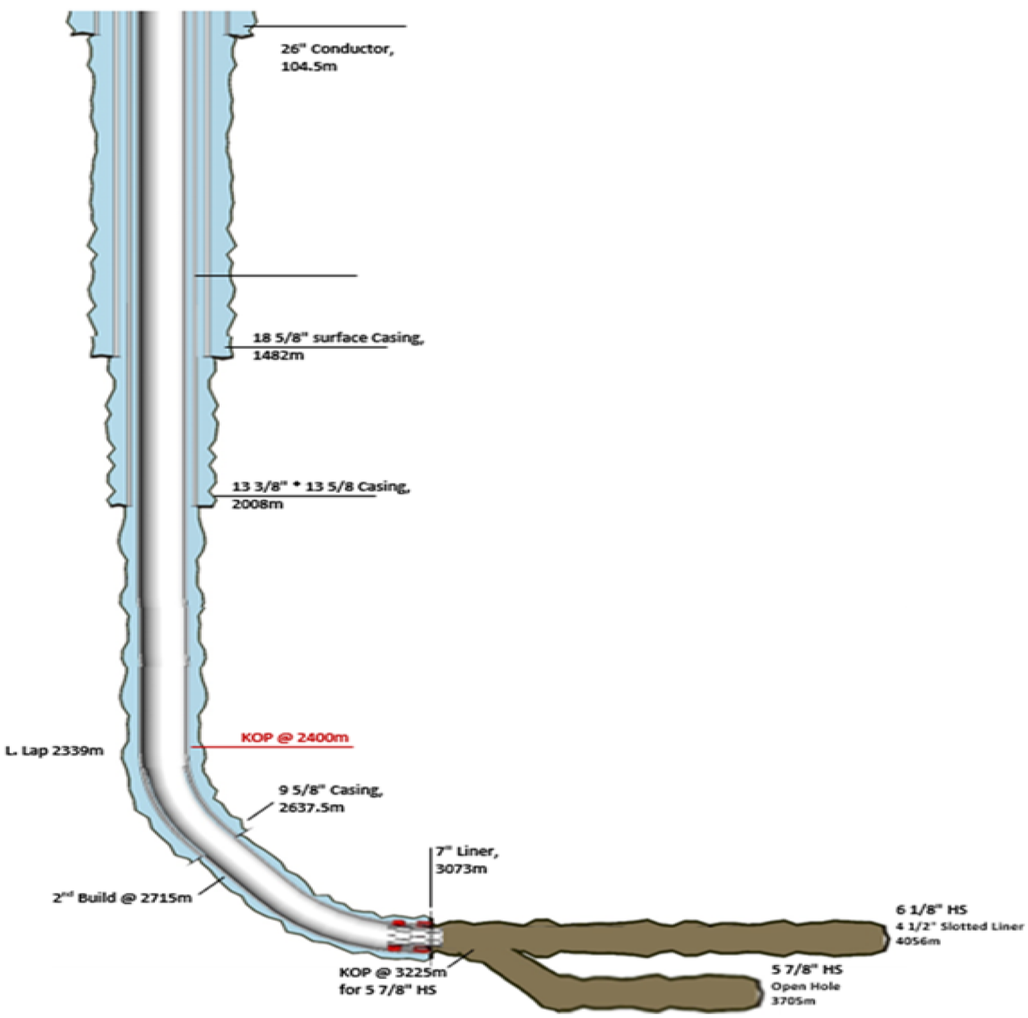


Figure 4—Well schematic of JR-09 showing dual-leg horizontal completion with open-hole sections in both the main and kick-off legs.

Stimulation Strategy & Methodology

The stimulation campaign for well X-1 was designed and executed under PECO's proprietary Planning and Execution (PECO-PE) framework, a comprehensive workflow tailored for challenging carbonate reservoirs with complex horizontal geometries. The planning process began with an extensive subsurface evaluation, integrating geological and structural modeling of the A carbonate reservoir with updated mapping and offset well data from JR-03, JR-04, and JR-05. These analyses helped refine the understanding of reservoir heterogeneity, natural fracture distribution, and the expected acid response. Image logs and core studies confirmed that the A-Main unit was a tight, micro-porous limestone with negligible matrix permeability and moderate heterogeneity. Reservoir fluids consisted of heavy oil with an API gravity of around 22°, high-salinity formation water exceeding 110,000 ppm, and a bottom-hole temperature of 209°F, all of which underscored the need for an advanced and carefully engineered acidizing solution.

In parallel, laboratory testing was performed to design a tailored acid system for the well. Compatibility studies and reactivity tests focused on 15% hydrochloric acid (HCl) and viscoelastic diverting acid (VDA), assessing their performance under actual reservoir conditions. The experiments evaluated fluid stability at reservoir temperature and salinity, gel breaking characteristics, and the ability of different overflushes to achieve clean backflow. Based on these results, a treatment recipe was developed that combined the deep penetration capability of HCl with the diversion efficiency of VDA, supported by carefully designed preflush and overflush stages to minimize the risk of post-treatment damage and ensure complete cleanup.

Once the treatment objectives were defined, engineering simulations were carried out to validate the design and guide operational parameters. Hydraulic models calculated friction losses along the coiled tubing string during selective acidizing, while force analyses confirmed the inflatable packer could be set and retrieved safely under expected pressure conditions. Acid placement simulations provided forecasts of wormhole propagation, diversion efficiency, and breakthrough risks, enabling optimization of injection rates, volumes, and staging to balance deep penetration with operational safety. Due to software limitations in directly modeling a dual-leg horizontal well, the acid placement simulations were performed using a single-leg configuration with an equivalent interval length representing both laterals. This approach allowed accurate calculation of design volumes and penetration profiles within the constraints of the modeling software. For production deliverability forecasting, however, an innovative dual-leg model was created in Multi phase flow software, enabling two-leg performance to be analyzed realistically and ensuring that both the selective acidizing and matrix stages were history-matched to field behavior.

Special attention was given to the design of the bottomhole assembly and the fluid placement strategy. The BHA incorporated a retrievable inflatable packer, check valves, a circulation sub, and an emergency release mechanism to enable selective acidizing of each leg independently. In addition, a kick-off string was engineered to deliver acid during the matrix acidizing phase. The decision to use a kick-off string and adopt a packer-less approach for this phase was the result of detailed engineering review. The available inflatable packer had a 3-inch element size, which exceeded the minimum ID of the intended packer setting zones. Because of this dimensional constraint, a conventional flowhead could not be used, and the wellhead itself became the operational interface for pressure control. This configuration was operationally sound but introduced several challenges during cleanup, including pressure imbalance between annulus and tubing, water accumulation behind the tubing, and extended cleanup times. These issues were managed effectively through periodic reverse circulation, which restored hydraulic balance and allowed the operation to progress safely.

Before execution, a Hazard Analysis and Risk Control (HARC) session was conducted with full participation from reservoir, operations, and safety teams. All potential risks, including packer inflation malfunction, coiled tubing buckling, acid misplacement, and unexpected pressure buildup, were analyzed, and a comprehensive contingency plan was developed. The fallback scenarios included bullhead acidizing, nitrogen lift support for cleanup, and emergency protocols for well control and pressure management.

The stimulation itself was implemented in a carefully sequenced workflow. Each horizontal leg was first treated selectively using coiled tubing-conveyed inflatable packers, which allowed precise placement of acid and ensured that damage in near-wellbore zones was addressed without fluid loss to unintended intervals. The main leg was treated with 290 barrels of 15% HCl, buffered by 195 barrels of pre- and overflush, while the kick-off leg received 196 barrels of acid with 98 barrels of bracketing fluids. Following this, a full matrix acidizing treatment was pumped through the kick-off string, consisting of 1,172 barrels of 15% HCl, 586 barrels of VDA, and an equal volume of brine overflush. Simulation forecasts had indicated that this treatment would generate 25–30 inches of wormhole penetration, opening deep flow channels and connecting previously unstimulated rock to the wellbore.

Once pumping operations were completed, the coiled tubing assembly was retrieved, and a structured well testing phase began to evaluate the impact of the stimulation. Multi-choke flow measurements, injectivity testing, and real-time pressure monitoring confirmed that the design objectives had been met. Throughout all stages, operational adjustments were made dynamically to ensure fluid placement integrity and maintain safe conditions at the surface and in the wellbore.

The successful execution of this stimulation, combining mechanical isolation during the selective phase with chemical diversion during the matrix stage, demonstrates how an integrated, simulation-driven design and a disciplined operational approach can transform the productivity of complex dual-leg open-hole carbonate wells. This methodology now provides a proven framework for future stimulations in the A reservoir and other tight carbonate systems.

Results and Discussion

Engineering / Design

Acid Stimulation Software simulation. To ensure the technical success of the dual-phase acidizing campaign in well X-1, a comprehensive set of acid placement simulations was performed using Acid Stimulation Software . Since it was unclear during the design phase which horizontal leg the coiled tubing (CT) would ultimately access in the field, simulations were conducted for two scenarios — CT fixed at a constant depth and CT progressively moving along the lateral — to ensure optimal execution under either operational condition.

Additionally, sensitivity analyses were carried out by varying the horizontal-to-vertical permeability ratio (kh/kv) and increasing the acid volume beyond the design values. These scenarios aimed to assess the impact of key reservoir and design parameters on acid penetration behavior and treatment uniformity.

In the selective acidizing phase, simulations demonstrated that progressive CT movement resulted in more uniform acid distribution and deeper penetration into damaged zones compared to a fixed-depth CT. In the main leg (3,073–4,056.6 m MD), approximately 390 bbl of 15% HCl was injected, buffered with about 195 bbl each of preflush and overflush. In the kick-off leg (3,225–3,705 m MD), approximately 196 bbl of acid and 98 bbl each of preflush and overflush were used. Comparing the two CT deployment scenarios showed that progressive CT movement achieved penetration depths of up to ~25 inches, while the fixed-depth scenario resulted in only ~18–20 inches of penetration. Fig. 5 clearly illustrates this finding.

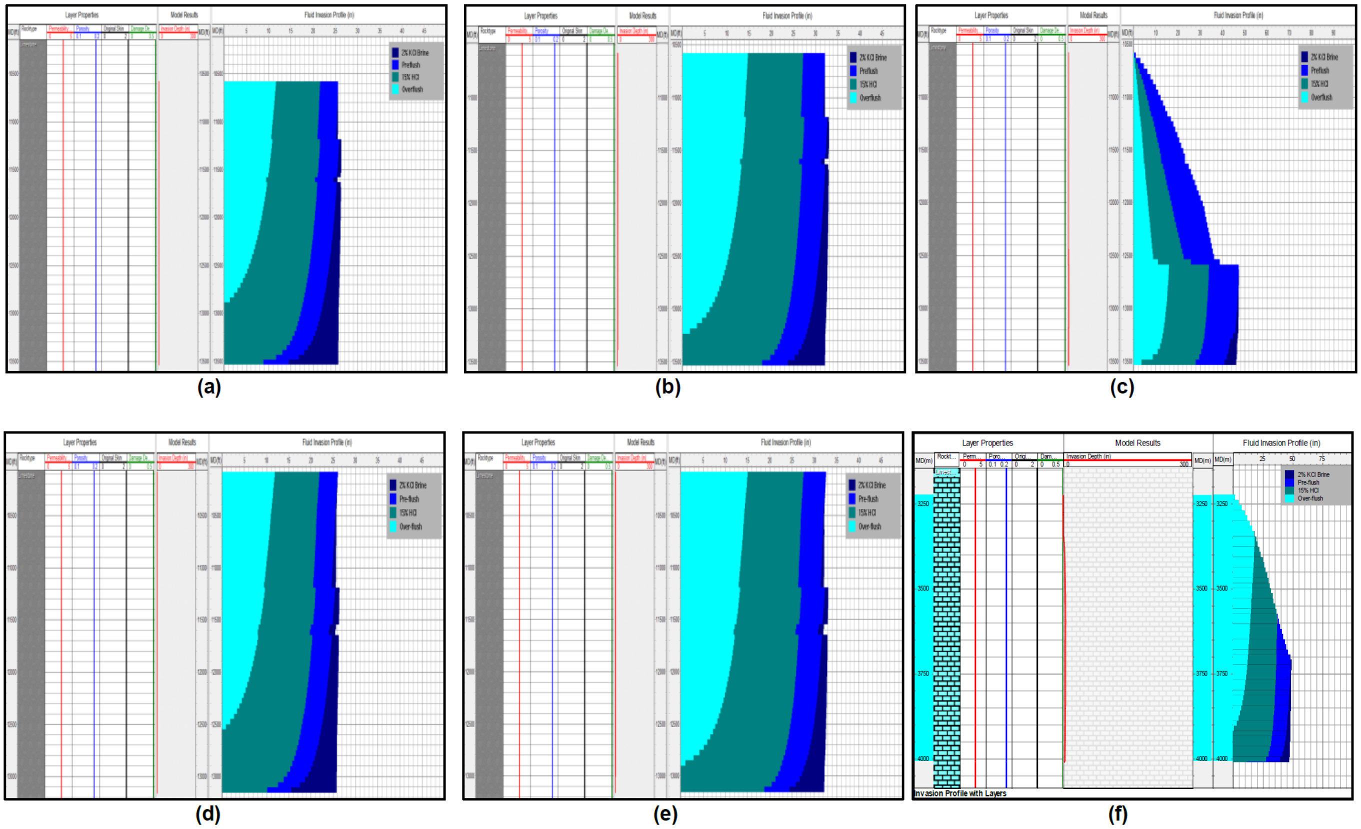


Figure 5—Simulated acid invasion profiles for selective acidizing scenarios in both laterals of X-1. (a) Fixed CT position in Leg 1; (b) Fixed CT position in Leg 1 with increased acid volume; (c) Progressive CT movement in Leg 1; (d) Fixed CT position in Leg 2; (e) Fixed CT position in Leg 2 with increased acid volume; (f) Progressive CT movement in Leg 2. Acid Stimulation Software results demonstrate that CT movement enabled more uniform acid coverage and deeper penetration (#25 inches) compared to fixed-depth injection (#18–20 inches). Increasing acid volume in fixed scenarios extended penetration but reduced treatment uniformity, while higher kh/kv values also led to preferential acid flow along dominant streaks.

Sensitivity analysis on acid volume revealed that increasing the acid beyond the designed amount increased penetration depth up to ~30 inches but at the expense of treatment uniformity. Larger volumes led to channeling along high-permeability streaks, elevated operational pressures, and a higher risk of premature fracturing. The analysis confirmed that the design volume provided the best balance between penetration, uniformity, and operational safety.

Varying the horizontal-to-vertical permeability ratio showed that higher kh/kv (dominant horizontal permeability) caused acid to flow preferentially along the lateral, reducing vertical penetration and treatment uniformity. Conversely, lower kh/kv values resulted in deeper and more uniform acid penetration into the formation, emphasizing the importance of understanding reservoir anisotropy when designing acidizing treatments.

In the matrix acidizing phase, approximately 1,172 bbl of 15% HCl, 596 bbl of VDA15, 586 bbl of overflush, and 293 bbl of preflush were injected in nine alternating acid and diverter stages. The simulation predicted a reduction in skin factor from +2.61 to −1.54. Fig. 6 shows the skin reduction trend over time, with a rapid drop during the first three acid-diverter cycles and stabilization at negative values by the end of the treatment (Step 9 Overflush). Sensitivity analysis revealed that increasing the acid volume beyond the design in this phase did not yield significant additional skin reduction but did result in higher operational pressures.

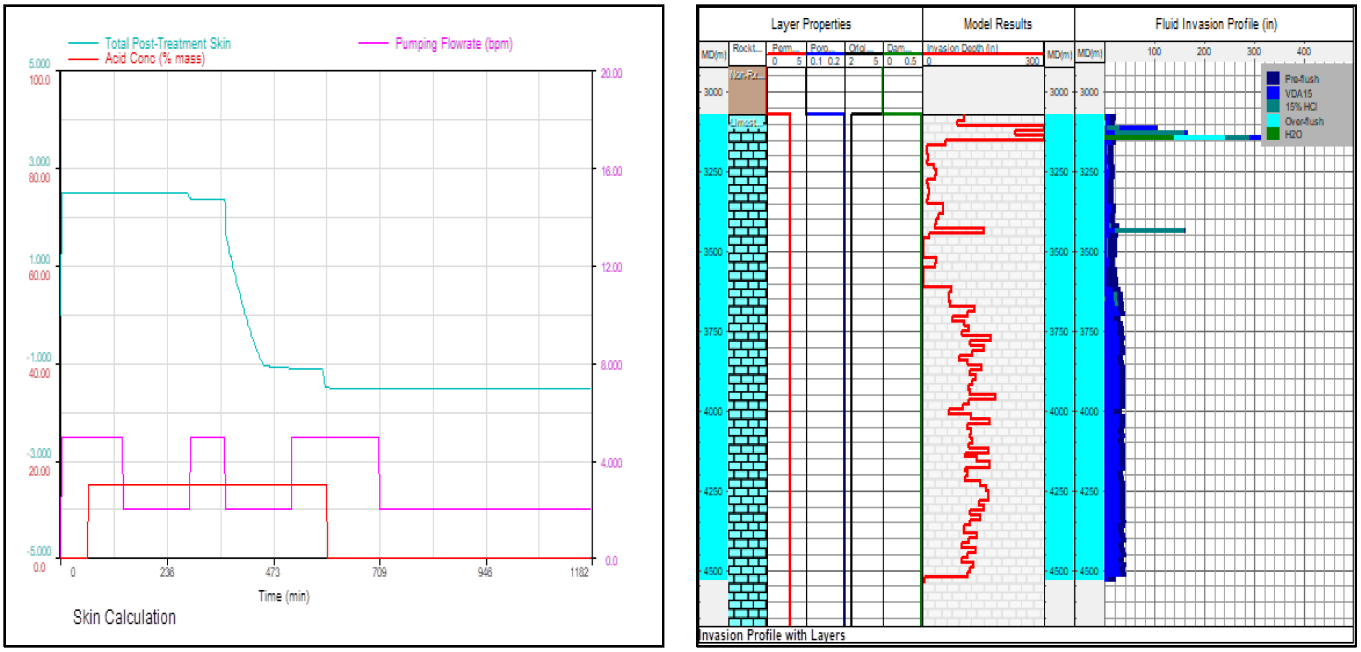


Figure 6—Calculated skin factor evolution and invasion profile during matrix acidizing. Left: Acid Stimulation Software simulation shows skin factor decreasing from +2.61 to −1.54 during the nine-stage treatment, with the sharpest reduction occurring in the early acid–diverter cycles, confirming complete removal of near-wellbore damage. Right: Invasion profile and layer properties indicate acid penetration of 200–280 inches in most layers, with select intervals exceeding 400 inches, demonstrating effective wormhole propagation across the A carbonate formation.

These findings clearly demonstrate that the innovative two-step acidizing strategy — selective acidizing through mechanical isolation followed by matrix acidizing with chemical diversion — provides an optimal balance of penetration depth, uniformity, efficiency, and safety. The Acid Stimulation Software simulations provided critical insights into optimal injection rates, effective volumes, and safe pressure envelopes, supporting real-time operational decisions and ultimately contributing to the success of the X-1 acidizing campaign.

Coiled Tubing Hydraulic and Stress Analysis Simulator. To complement the Acid Stimulation Software acid placement studies, a detailed coiled tubing mechanical simulation was performed to evaluate the mechanical integrity of the coiled tubing (CT) string and bottomhole assembly (BHA) throughout all operational phases of the X-1 dual-leg stimulation. The analysis assessed surface weight behavior, slack-off and overpull margins, and potential risks of lock-up, buckling, or tensile failure during displacement, nitrogen lifting, and stimulation.

The Coiled Tubing Hydraulic and Stress Analysis Simulator model incorporated the complete CT string – approximately 4,213 m in length including the BHA – and extended to the target inflatable packer setting depth of 4,056.6 m. Key inputs such as tool weight, borehole friction coefficients, drag forces, and dynamic hydraulic effects were integrated to ensure a realistic mechanical forecast.

As illustrated in Fig. 7, the surface weight exhibited a smooth and predictable increase with depth during run-in-hole (RIH) and pull-out-of-hole (POOH). The model predicted a lock-up depth of 3,901.4 m, yet the CT string successfully reached the full target depth without incident. The calculated slack-off force (2,502 lbf) and maximum overpull (5,538 lbf) were significantly below the tubing's yield and buckling limits, providing a substantial mechanical safety margin. Correspondingly, Fig. 8 confirms that circulation and wellhead pressure (WHP) remained stable throughout displacement, validating the hydraulic stability of the operation.

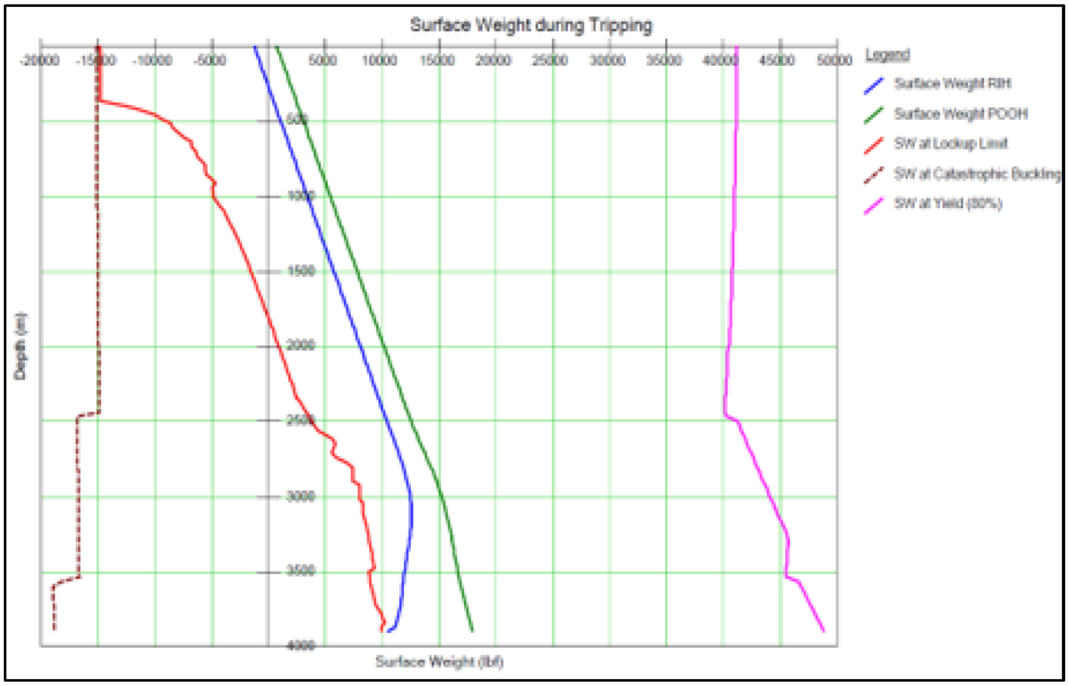


Figure 7—Modeled surface weight vs. depth during displacement phase. Lock-up depth conservatively predicted at 3,901.4 m, yet CT successfully advanced to target depth.

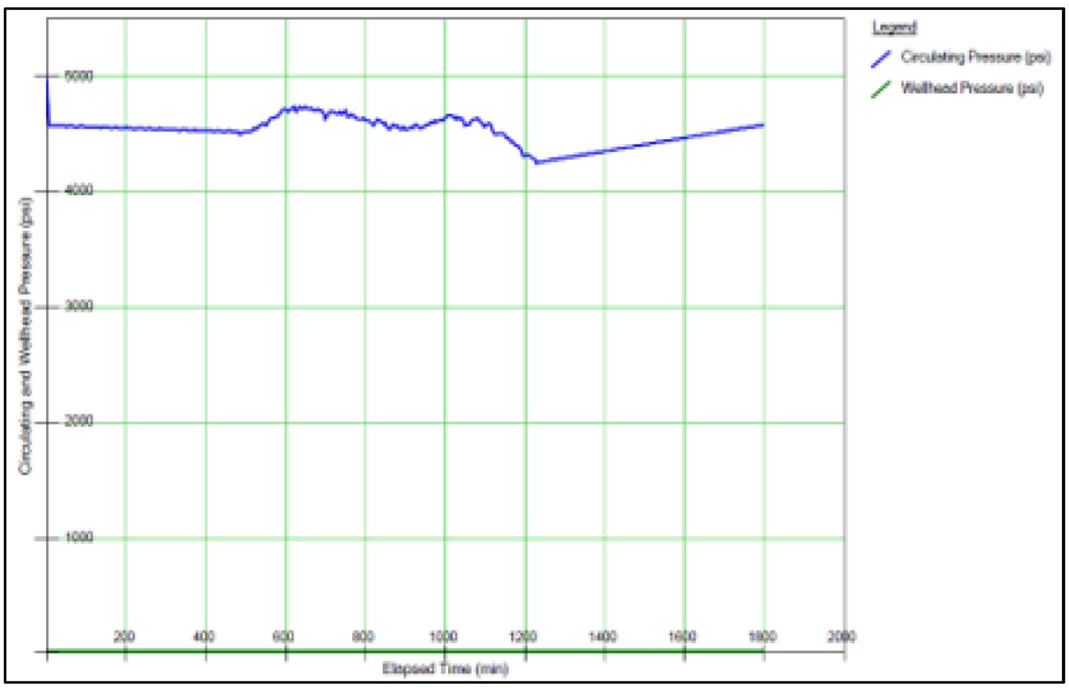


Figure 8—Simulated circulation rate and wellhead pressure during displacement, confirming stable hydraulic conditions.

During the N_2 lifting stage, drag behavior and weight response remained consistent with the displacement phase. As shown in Fig. 9, the surface weight trend displayed no abnormal fluctuations, and the CT string advanced without any signs of instability. The modeled WHP, illustrated in Fig. 10, stayed within the planned operating window of 4,200–4,700 psi, confirming that nitrogen circulation would not impose undue mechanical strain or risk of tubing movement.

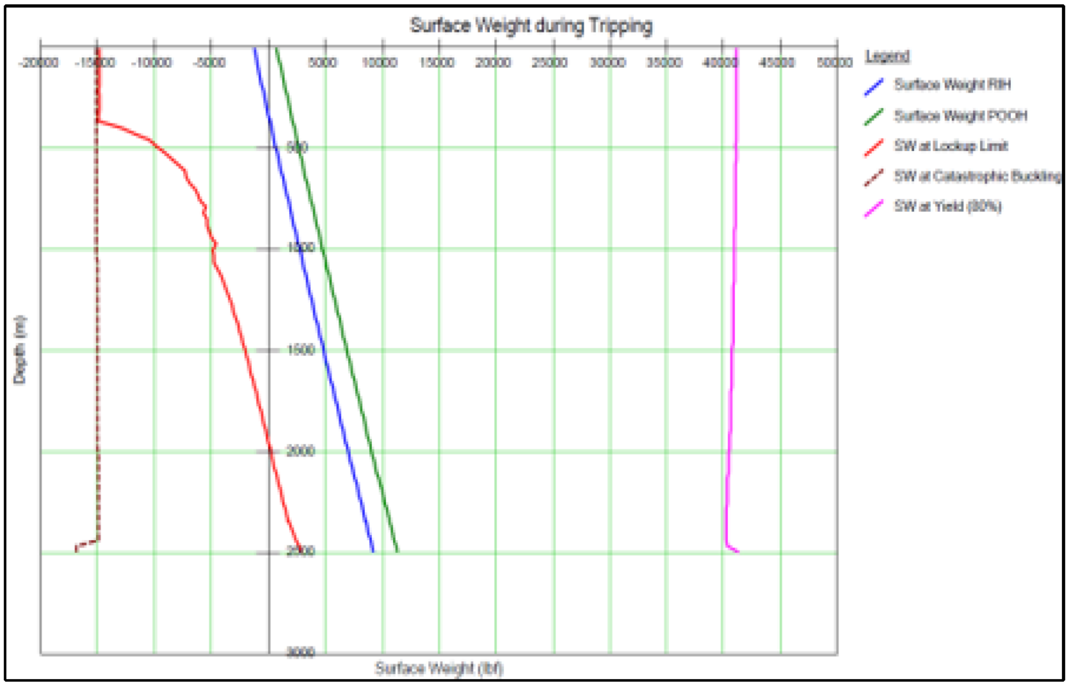


Figure 9—Surface weight behavior during nitrogen lifting showing consistent drag trends and absence of instability.

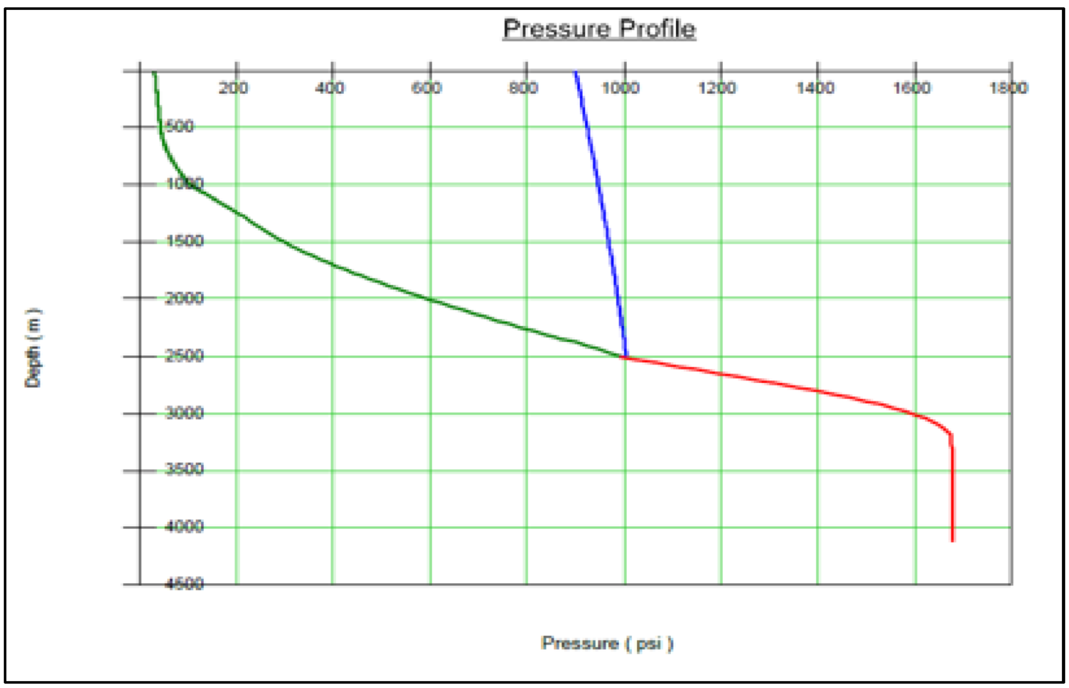


Figure 10—Pressure profile during nitrogen lifting stage demonstrating WHP stability within 4,200–4,700 psi range.

The stimulation phase represented the most mechanically demanding period due to fluid density changes and elevated circulation pressures. Coiled Tubing Hydraulic and Stress Analysis Simulator modeling forecasted a slightly shallower lock-up depth (3,829.7 m) compared to displacement; however, optimized slack-off management and WHP control allowed safe CT deployment to the full 4,056.6 m depth. As seen in Fig. 11, the surface weight remained within expected limits, and no signs of buckling or tool sticking were detected. Fig. 12 further demonstrates that circulation and WHP variations during acid and diverter injection were well-controlled and remained within design tolerances.

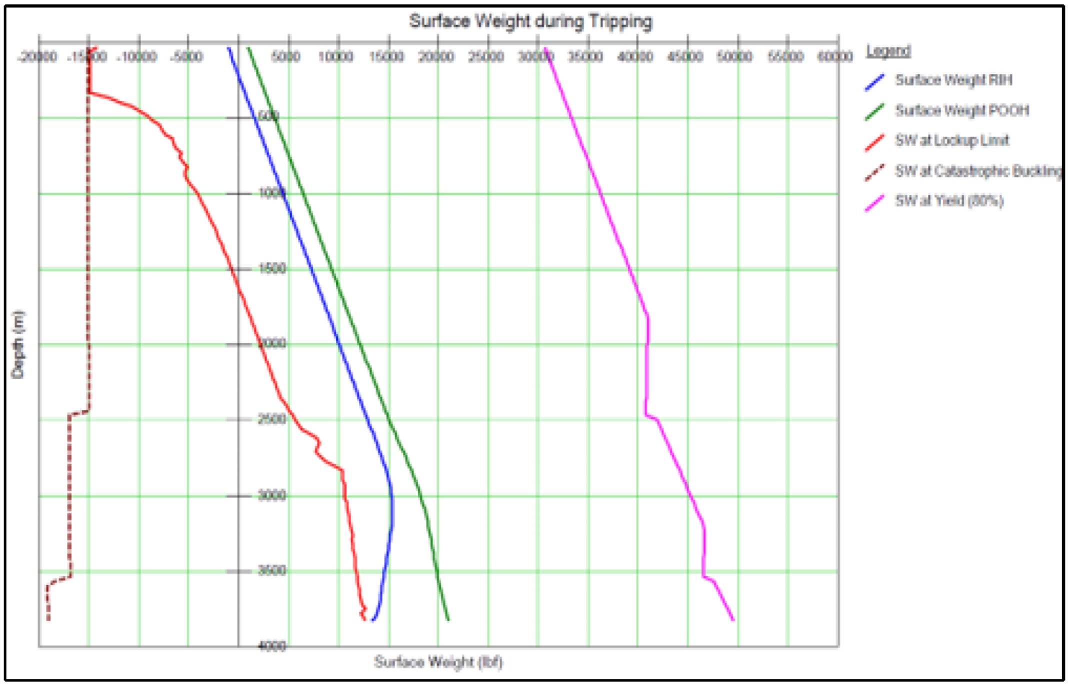


Figure 11—Surface weight vs. depth during stimulation phase. Lock-up depth modeled at 3,829.7 m with CT safely reaching 4,056.6 m.

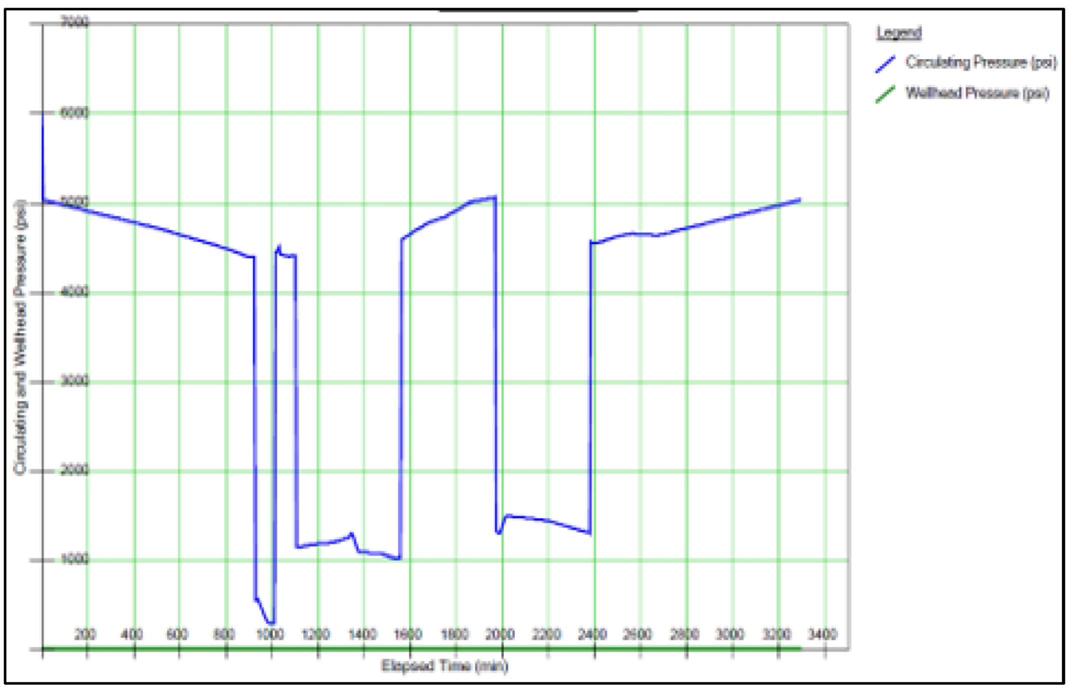


Figure 12—Simulated circulation and wellhead pressure during stimulation, showing controlled transients despite fluid density changes.

Across all scenarios, Coiled Tubing Hydraulic and Stress Analysis Simulator confirmed that the CT string could be deployed and retrieved with ample safety margin. The worst-case overpull of 5,538 lbf represented only ~25% of the CT's rated tensile capacity, underscoring the robustness of the design. Although lock-up depths were conservatively predicted to be 150–250 m shallower than the operational target, application of dynamic slack-off techniques and friction management ensured full-depth access.

Beyond validating the mechanical design, Coiled Tubing Hydraulic and Stress Analysis Simulator modeling was pivotal in defining a safe operational window for slack-off, circulation, and packer inflation. It also informed contingency measures, including potential nitrogen assistance and fluid viscosity adjustments. This analysis provided the mechanical assurance necessary for safe execution and directly contributed to the success of the X-1 dual-leg stimulation.

Inflatable packer analysis. The selection and deployment of the inflatable packer in well X-1 was a cornerstone of the dual-leg stimulation strategy and directly influenced the overall success of the operation. Given the open-hole completion geometry of the A carbonate reservoir — comprising 6.125-inch and 5.875-inch hole sections in the main and kick-off laterals respectively — achieving precise zonal isolation was critical to avoid cross-flow between laterals and ensure selective acid placement. After evaluating conventional options such as straddle packers and mechanical isolation tools, the engineering team determined that these alternatives either posed a risk of mechanical sticking or could not reliably navigate the 1.5-inch coiled tubing (CT) ID. Consequently, a thru-tubing inflatable packer was selected as the only viable solution to deliver effective isolation while maintaining operational flexibility.

A comprehensive validation program supported this selection. Laboratory testing was conducted under simulated wellbore conditions reflecting X-1's bottom-hole temperature (209 °F), salinity (>110,000 ppm), and expected pressure envelope. Tests confirmed the packer's elastomer integrity, expansion capability, and sealing efficiency across the anticipated borehole diameters. In addition, a field validation step was performed: prior to operations, activation balls were circulated through the CT to verify unobstructed passage and confirm the packer's ball-seat actuation mechanism would function flawlessly in live conditions.

Caliper log analysis identified the most dimensionally stable zone at 3,250 m MD, which was selected as the primary setting depth. At this interval, the borehole diameter aligned with the packer's optimal expansion range, ensuring a reliable seal without overstressing the element. The engineering team referenced the performance matrix (Fig. 13) to confirm that the selected 3.0-inch element could safely expand to seal both 5.875-inch and 6.125-inch open-hole sections while providing a differential pressure rating of 4,500–5,000 psi — well above the maximum treatment pressure of ~3,000 psi observed during acidizing. Burst and collapse ratings offered additional safety margin, further mitigating the risk of mechanical failure under high differential loads.

Element OD Size (in.)	3.00
Max. Setting Depth ID (in.)	6.125 Open-Hole
Element Service Type	Acid
Element Pressure Rating (psi)	3600

Element OD [inch]	ID of Casing [inch]													
	2.441	2.992	3.548	3.958	4.151	4.892	5.921	6.094	6.765	7.511	8.681	9.760	10.772	12.415
	Max Differential Pressure [psi]													
1.690	5000	4000	3000	2300	2000	1200								
2.125	5500	5100	4400	3900	3600	2600	1600	1400						
2.500		6500	6000	5500	5200	4200	2600	2300	1600					
2.750		7000	6600	6200	6000	5000	3500	3200	2300	1600				
3.000			8000	7500	7300	6500	4800	4500	3400	2500	1500			
3.375				8500	8200	7500	6000	5600	4500	3500	2300	1500		
4.250						8500	6800	6500	5500	4500	3000	2000		
5.375							8500	8200	7500	6800	5300	4000	3200	2000
6.500										8000	6500	5000	3800	2500

Figure 13—Thru-Tubing Inflatable Element Pressure Rating.

The inflation procedure was executed under tight hydraulic control, consistent with pre-job simulations and risk assessments. Inflation pressure was gradually increased to 600 psi to shear the delayed-inflate screws, then held at 1,000 psi for five minutes to allow the element to conform to the borehole wall. Pressure was incrementally raised in 500-psi steps up to 2,100 psi, where the activation ball seat sheared, confirming full inflation. For deflation, a controlled overpull of 5,250 lbf was applied via the CT-conveyed kick-off string to shear the release sleeve, followed by a 20-minute waiting period for full element collapse before retrieval.

Integration of the inflatable packer into the CT bottomhole assembly (BHA) (Table 1) was another critical engineering step. The BHA — comprising a dimple connector, dual flapper check valve, emergency disconnect, circulation sub, and the packer body itself — was designed to maintain both pressure integrity and emergency release capability. This design ensured that, in the event of packer malfunction, the CT could be safely retrieved without compromising well control.

Table 1—CT Bottomhole Assembly while acidizing with inflatable packer

Item	Tools/ Component Description	Max. Tool OD, in	Min. Tool ID, in	Tool Length, in	Accu. BHA Length, in	Top Thread	Bottom Thread
1	Dimple Connector	2.125	NA	8.0	8.0	-	1.5" AMMT Pin
2	Double Flapper Check Valve	2.125	-	10.22	18.2	1.5" AMMT Box	1.5" AMMT Pin
3	Emergency Disconnect	2.125	0.65	12.8	31	-	1.5" AMMT Box
4	Circulation sub	2.125	0.55	9.6	40.6	-	-
5	Body of packer	3	0.39	118.8	159.4	-	-
Total Length of BHA, in (m):					159.4" (4.05 m)		

From an engineering and operational standpoint, the use of an inflatable packer enabled precise leg-by-leg acid placement during the selective acidizing phase, ensuring damaged zones were effectively treated without fluid loss to the adjacent lateral. This not only improved stimulation efficiency but also reduced chemical volumes and operational risk compared to a bullhead-only approach. The robust testing and validation program — spanning laboratory conditions, field ball-pass confirmation, and pre-job hydraulic modeling — provided the assurance that the packer would perform reliably, which it did: no leakage, lock-up, or retrieval issues were encountered.

Ultimately, the inflatable packer's performance was pivotal to the success of X-1's dual-leg stimulation, establishing a benchmark for future open-hole carbonate applications in the region.

The operational viability of the inflatable packer was further assessed using detailed manufacturer-provided performance charts and hydraulic response envelopes (Fig. 14 and Fig. 15). These figures offer a visual representation of the packer's pressure-handling limits and expansion behavior under actual field conditions.

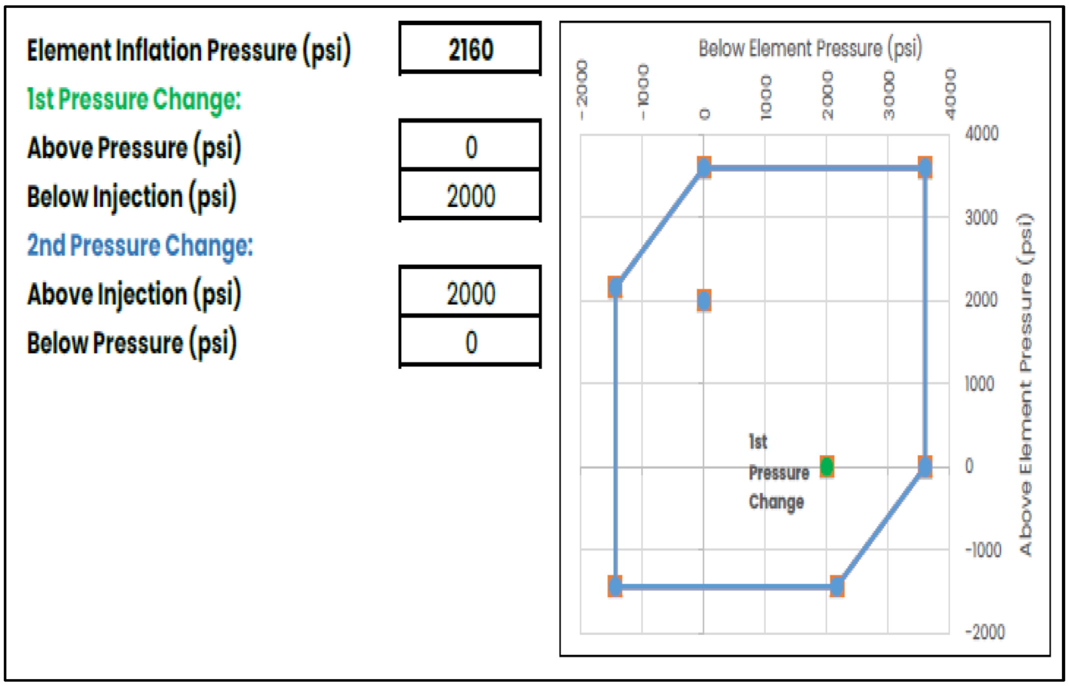
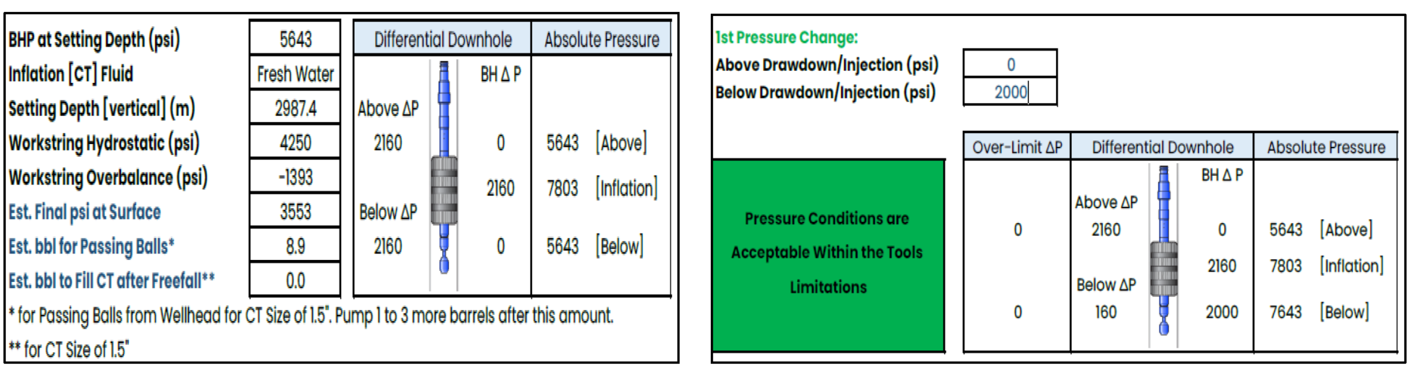


Figure 14—Thru-Tubing Inflatable Element Performance Envelope



(a) (b)

Figure 15—Thru-Tubing Inflatable Element Hydraulics: (a) Initial Condition (b) Final Condition

As shown in Fig. 14, the pressure envelope of the 3.0-inch thru-tubing inflatable element confirms its suitability for sealing both 5.875-inch and 6.125-inch open-hole sections. The envelope demonstrates a safe operating window of up to 5,000 psi differential pressure, comfortably exceeding the maximum observed treatment pressure of ~3,000 psi during acidizing operations. This margin highlights the element's robustness under combined mechanical and thermal stress, ensuring full circumferential contact without risking element rupture or collapse.

Fig. 15(a) and Fig. 15(b) illustrate the hydraulic transition of the inflatable element from its initial pre-inflation state to its final expanded configuration. The hydraulic profile confirms that the element achieved full expansion within the design pressure range (600–2,100 psi), validating the staged pressurization protocol described earlier. Moreover, the pressure-volume behavior confirms that expansion was uniform and progressive, minimizing the risk of localized overstressing or ballooning in irregular borehole geometries.

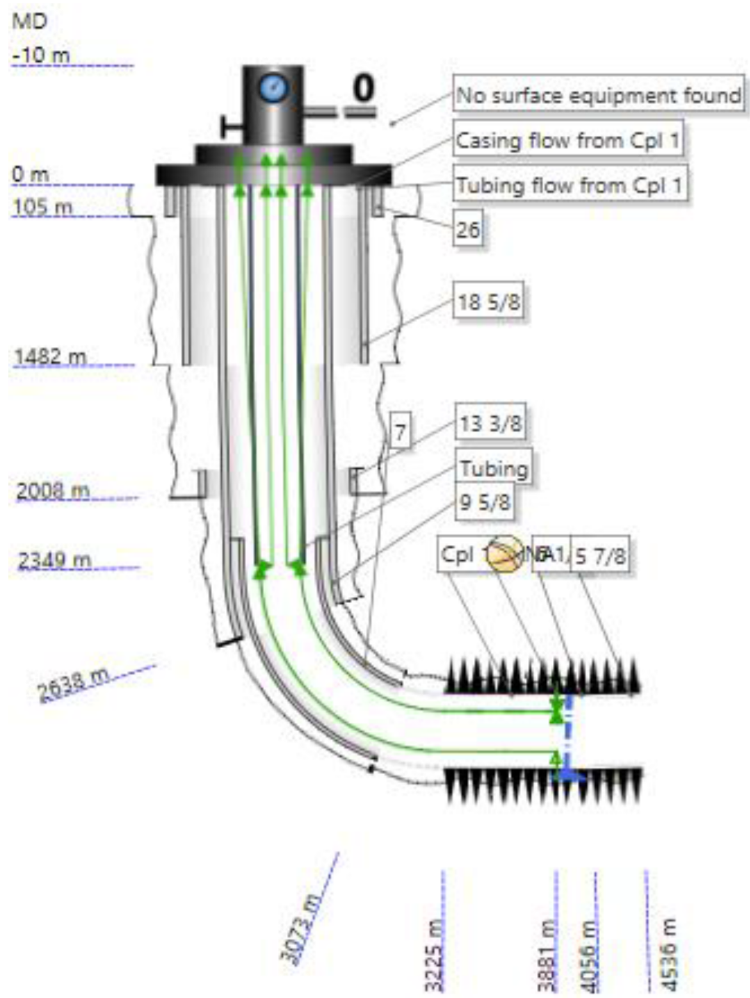


Figure 15a—Multi phase flow software well model of X-1

These results reinforce the selection of the 3.0-inch element as a technically sound choice, combining mechanical adaptability, operational safety, and reliable sealing performance. Together with laboratory testing and field validation, the hydraulic performance envelope and pressure transition curves provided essential assurance of packer integrity and contributed directly to the safe and effective execution of the selective acidizing campaign in X-1.

Operational Execution and Assessment

Multi phase flow software Modeling. A comprehensive Multi phase flow software modeling study was conducted to evaluate the deliverability of well X-1 before and after stimulation, and to quantify the impact of the dual-phase acidizing strategy on inflow performance and skin factor reduction. The modeling workflow included baseline (pre-acid) calibration, post-selective acidizing updates, and post-matrix acidizing history matching, followed by a comparative interpretation of all three stages.

Pre-acid-1 well test modeling

Multi phase flow software modeling was first conducted to establish the baseline deliverability for well JR-09 before any stimulation work was performed. The model used full well schematic details and was calibrated against pre-stimulation well test data. Reservoir pressure was 5,670 psia, with reservoir temperature at 208 °F. At the measured pre-stimulation flow rate of 1,796 STB/d, the flowing bottomhole pressure (Pwf) was 3,780 psia, yielding a Productivity Index (PI) of only 0.95 STB/d/psi.

This PI value, along with observed well performance, confirmed significant near-wellbore damage and restricted inflow capacity. The baseline Inflow Performance Relationship (IPR) curve (Fig. 16) demonstrates this limitation: achieving even moderate production rates required substantial drawdown. For example, extrapolation of the IPR showed that at 3,500 STB/d, bottomhole pressure would approach 2,000 psia, indicating that higher production targets would demand impractically high drawdowns.

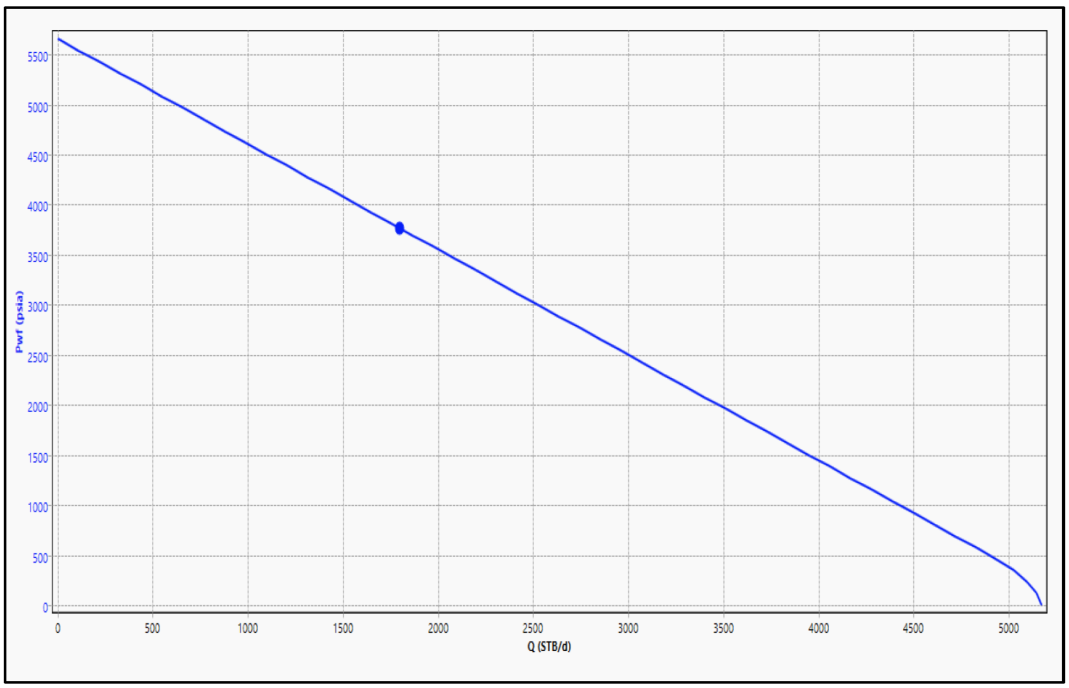


Figure 16—Baseline IPR curve for JR-09 before stimulation, showing limited inflow due to a skin factor of 4.5.

The pressure profile generated from the baseline model (Fig. 17) further illustrates the flow constraints in well JR-09 prior to stimulation. At reservoir depth, pressure starts at 5,670 psia, but by the time fluids reach the wellhead, the outlet pressure has dropped to just ~197 psia at the modeled rate.

This steep and continuous decline confirms that the majority of pressure losses originate from near-wellbore damage and frictional effects in the wellbore. The profile demonstrates that significant drawdown is required simply to lift fluids to the surface, even at relatively modest rates.

The temperature profile (Fig. 18) derived from the Multi phase flow software model provided additional insights into the well's thermal behavior prior to stimulation. The reservoir temperature of 208 °F gradually declined as fluids ascended the wellbore, reaching approximately 120 °F at the wellhead. This cooling trend was progressive, without sudden drops, indicating a stable thermal regime under steady-state production. However, the analysis of this temperature profile served as a basis for assessing thermal stability after acidizing, ensuring that increased production rates would not induce sudden temperature changes or thermal shocks to the formation or surface facilities.

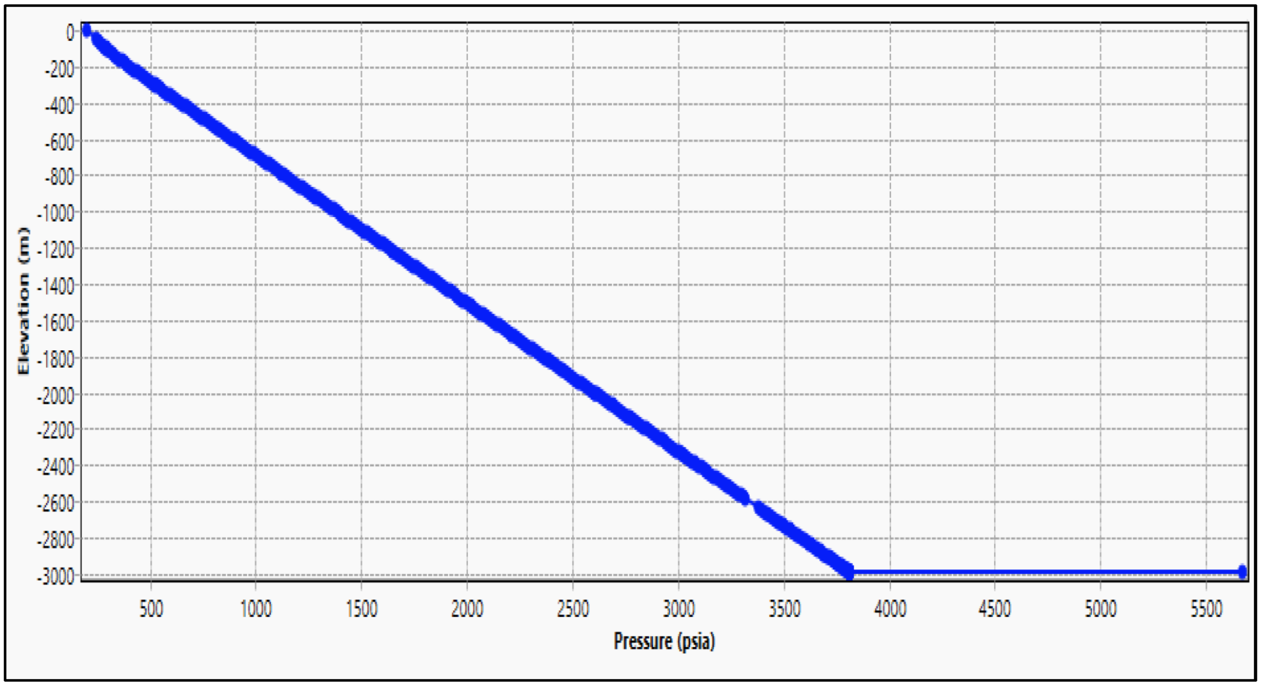


Figure 17—Pressure profile for JR-09 at various flow rates before stimulation, illustrating severe pressure losses.

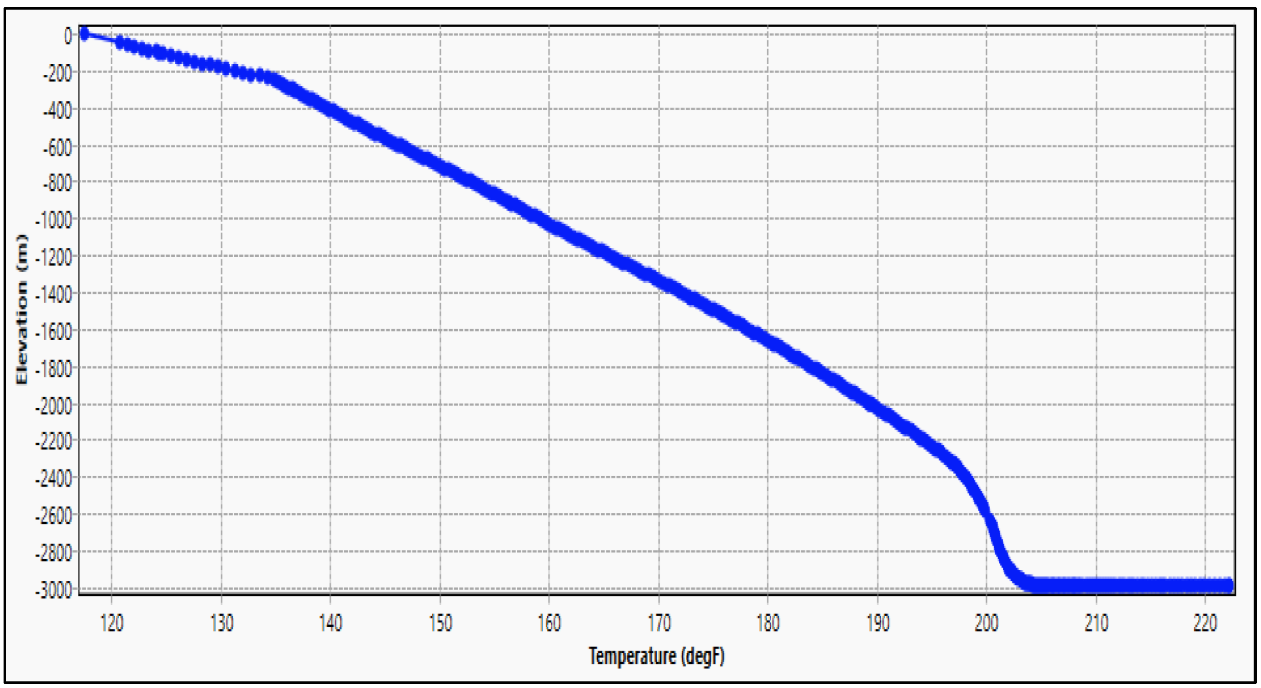


Figure 18—Baseline temperature profile for JR-09 prior to stimulation.

The wellhead pressure versus rate relationship (Table 2) clearly reflects the severe inflow restriction of JR-09 prior to stimulation. At modest rates of 500–1,000 STB/d, wellhead pressure remains relatively high (946–1,467 psia). However, as production approaches 1,500–2,000 STB/d, WHP collapses dramatically to less than 100 psia. By 2,500 STB/d, WHP drops to almost atmospheric levels (~12 psia), demonstrating that the well cannot sustain higher drawdowns safely without risking multiphase flow instability and operational hazards.

Table 2—Wellhead Pressure vs. Rate – Baseline Case

Rate (STB/day)	Wellhead Pressure (psia)
500	1467
1000	946
1500	427
2000	75
2500	12

To ensure the robustness of the Multi phase flow software baseline model, the calculated Inflow Performance Relationship (IPR) was rigorously history-matched against the theoretical Darcy IPR curve. The Darcy curve represents an ideal, damage-free reservoir condition, whereas the Multi phase flow software back-pressure IPR reflects actual well performance under pre-stimulation conditions.

Fig. 19 illustrates this comparison, showing that at lower flow rates (≤ 1,500 STB/d), the deviation between the two curves was moderate, but as flow rate increased, the gap widened significantly. This divergence is a direct indicator of additional pressure drop caused by near-wellbore damage and formation impairment.

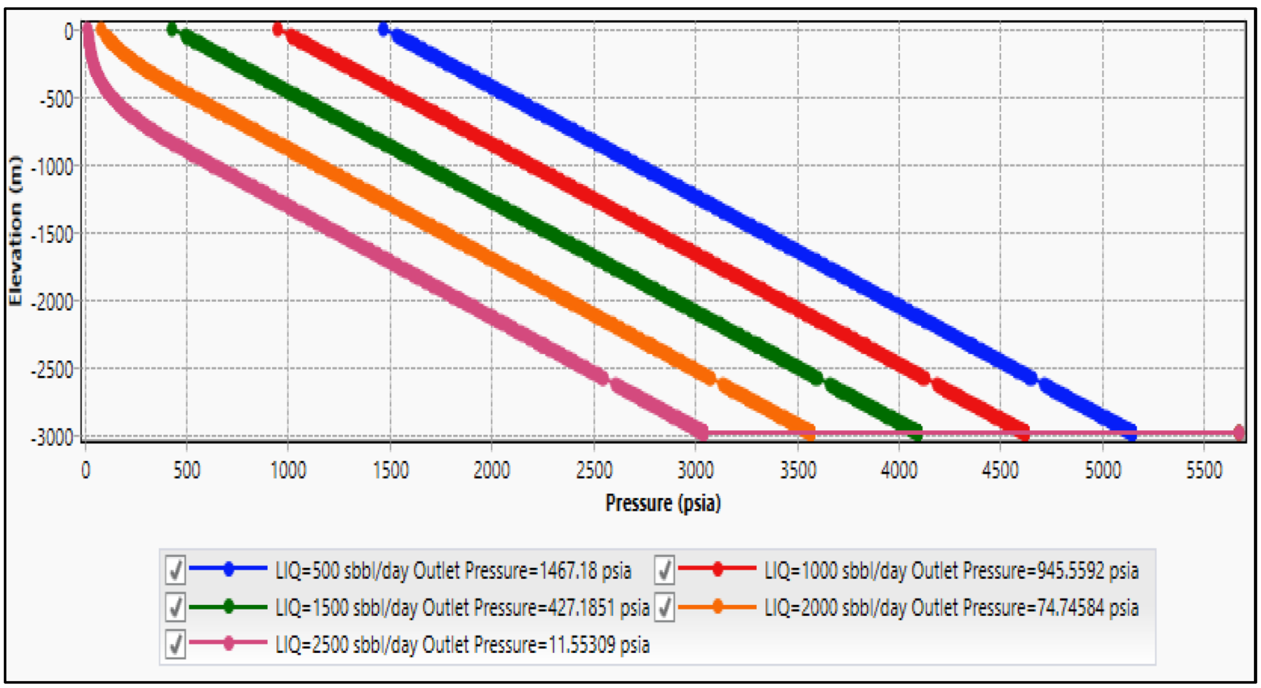


Figure 19—Wellhead pressure vs. rate plot for JR-09 before stimulation.

Through iterative adjustment of the Multi phase flow software model—primarily modifying skin factor parameters and validating against field test points—the back-pressure IPR curve was successfully aligned with the Darcy reference line at the lower-rate segment, while the residual offset at higher rates quantified the magnitude of the skin effect. The final calibration yielded a skin factor of 4.5, a value consistent with the pressure profile interpretation and well test analysis (Fig. 20).

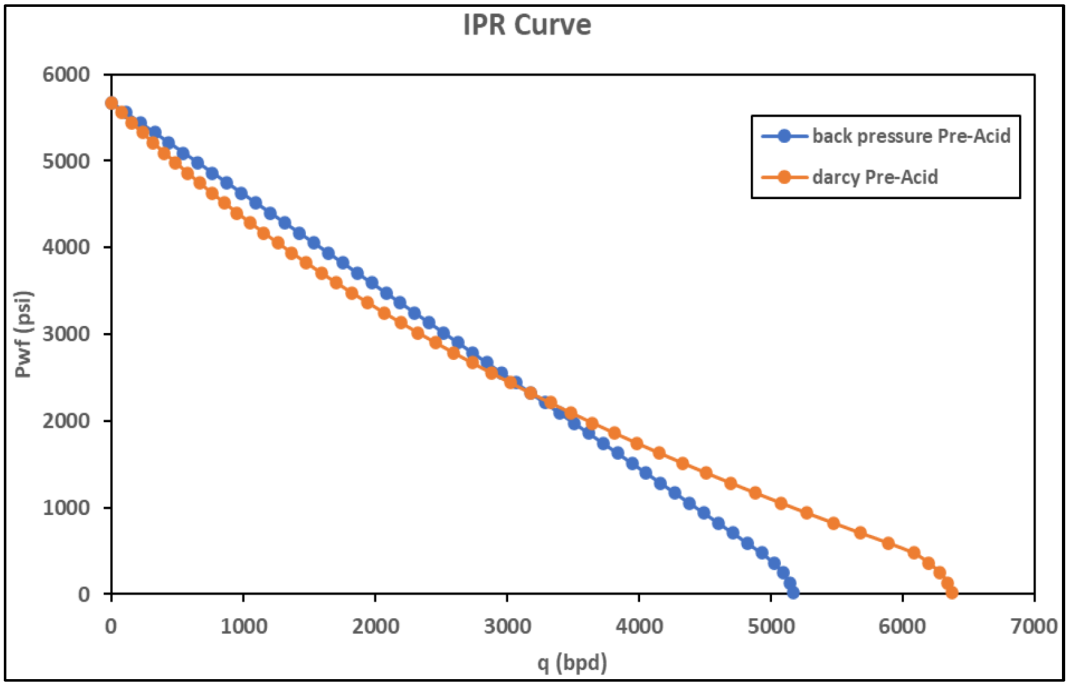


Figure 20—History match of Multi phase flow software back-pressure IPR with Darcy IPR confirming skin factor of 4.5.

From a reservoir engineering perspective, a skin of 4.5 implies: a significant reduction in effective wellbore radius, causing localized high-velocity flow zones; additional pressure losses on the order of several hundred psi, as evidenced by the divergence between modeled and idealized performance; and a clear justification for stimulation, since removal or reduction of this skin could unlock a major productivity gain. This history-matching exercise not only validated the Multi phase flow software model's accuracy but also established a quantified benchmark for evaluating post-stimulation improvements.

To validate the model calibration, a separate history match was also generated using an extended IPR curve across the full production range. However, since the available well test data and reservoir productivity for X-1 are concentrated in the lower-rate regime, the curve shown in Fig.20 — which provides a more accurate fit within this operational envelope — was ultimately selected as the representative match. The additional match is shown in Fig. 21, which covers a broader range but includes less reliable behavior in the early portion of the curve where field data are densest.

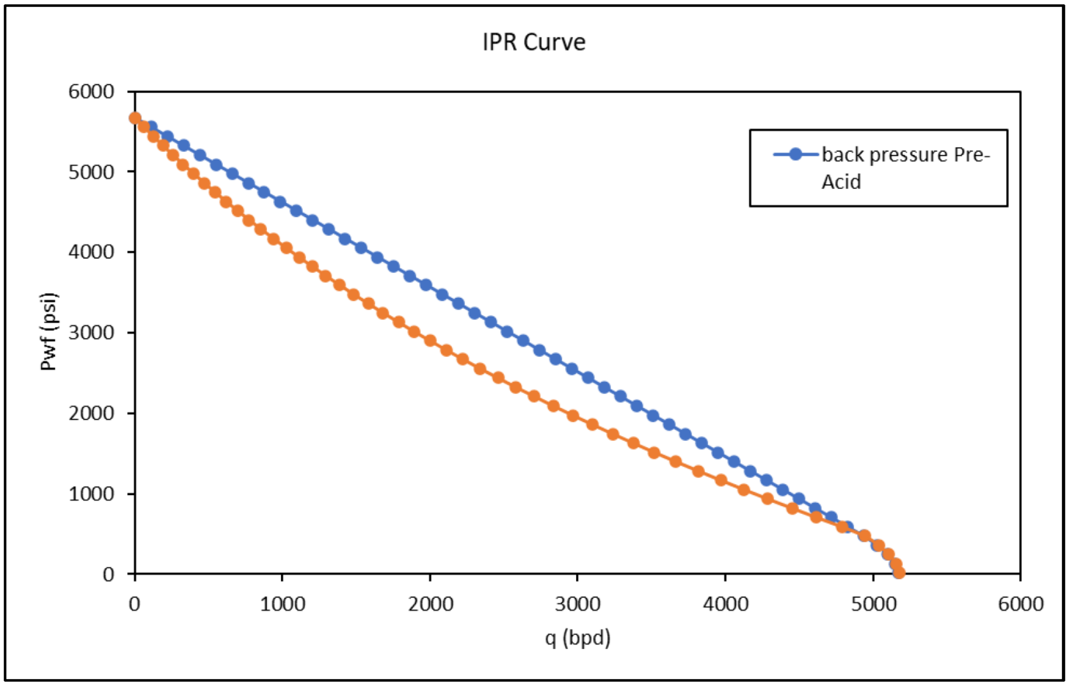


Figure 21—Extended IPR history match generated for well JR-09. Although this curve includes a wider flow range, it was not selected for final model calibration due to limited alignment with the well test points concentrated in the low-to-mid flow region.

Pre-acid-2 well test modeling

Following the selective acidizing phase, the Multi phase flow software model was recalibrated using updated well test data to evaluate the immediate impact of near-wellbore cleanup on well JR-09's performance. The post-acid test indicated measurable improvement: at an average stabilized flow rate of ~1,750 STB/d, the flowing bottomhole pressure (Pwf) was ~4,080 psia, resulting in a new Productivity Index (PI) of 1.10 STB/d/psi — a ~15% increase from the pre-acid PI of 0.95 STB/d/psi.

The revised Inflow Performance Relationship (IPR) curve (Fig. 22) illustrates the improved inflow. For any given drawdown, JR-09 could now sustain higher production rates, and the curve shifted closer to the theoretical Darcy line. A history-matching exercise between the back-pressure IPR and the Darcy reference (Fig. 23) confirmed that the skin factor was reduced from 4.5 to 3.0, demonstrating partial removal of near-wellbore damage while validating the Multi phase flow software model calibration.

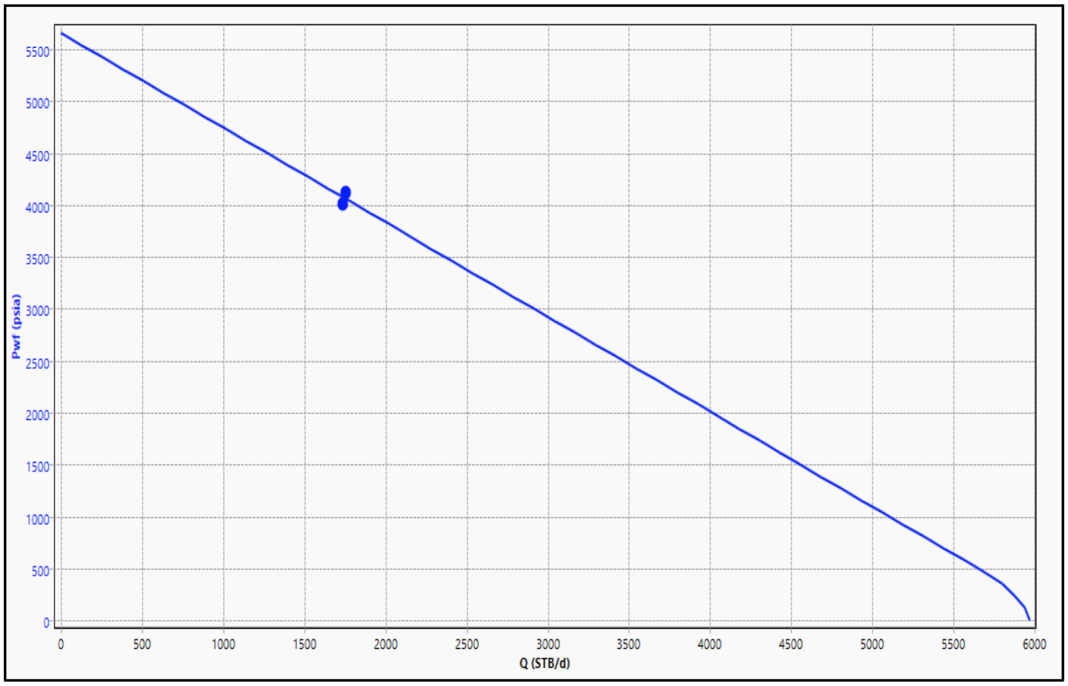


Figure 22—History match of back-pressure IPR

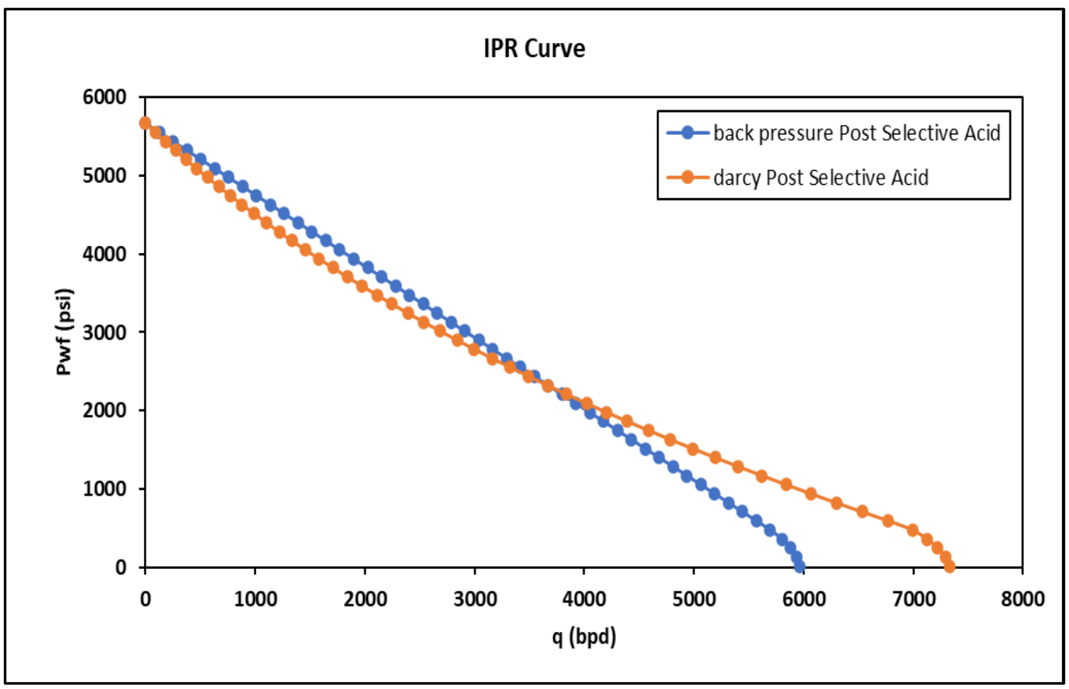


Figure 23—History match of back-pressure IPR with Darcy curve indicating reduction of skin factor from 4.5 to 3.0.

The post-acid pressure profile (Fig. 24) further highlights this improvement. Compared to the pre-acid profile (Fig. 17), the near-wellbore pressure drop was less steep, and pressure declined more evenly along the wellbore, reflecting improved permeability and reduced flow resistance in the damaged zone. Similarly, the updated temperature profile (Fig. 25) showed a smooth and progressive cooling trend from ~208 °F in the reservoir to ~122 °F at the wellhead, with no thermal anomalies, confirming tha…

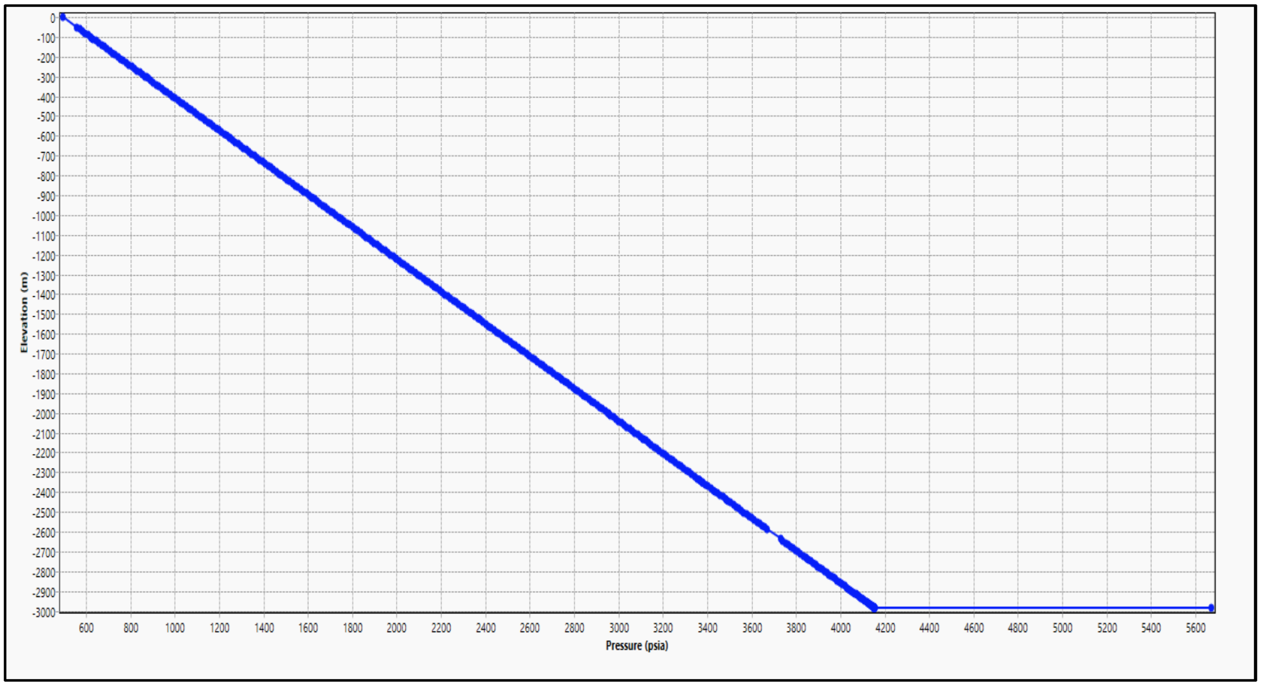


Figure 24—Post-acid pressure profile illustrating smoother gradient and reduced near-wellbore losses.

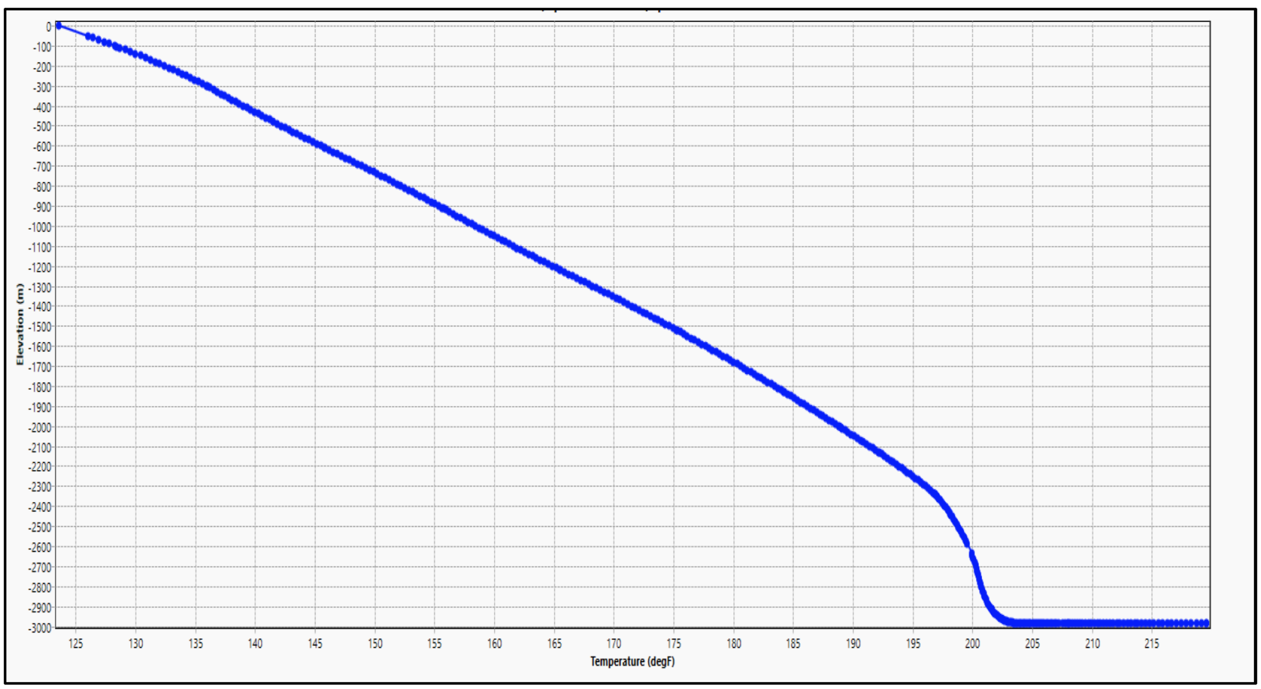


Figure 25—Post-acid temperature profile showing progressive and stable cooling trend.

The post-acid wellhead pressure versus rate relationship (Table 3 and Fig. 26) supports these findings. WHP remained significantly higher at equivalent production rates compared to the baseline case. At modest rates (1,000–1,500 STB/d), WHP held at 1,092–641 psia, whereas in the pre-acid model WHP collapsed more sharply. At rates above 2,500 STB/d, WHP again approached atmospheric levels, indicating that while selective acidizing improved deliverability, full restoration of the well's productive poten…

Table 3—Post-Selective Acidizing WHP vs. Rate

Rate (STB/d)	Wellhead Pressure (psia)
1000	1092
1500	641
2000	230
2500	44
3000	9

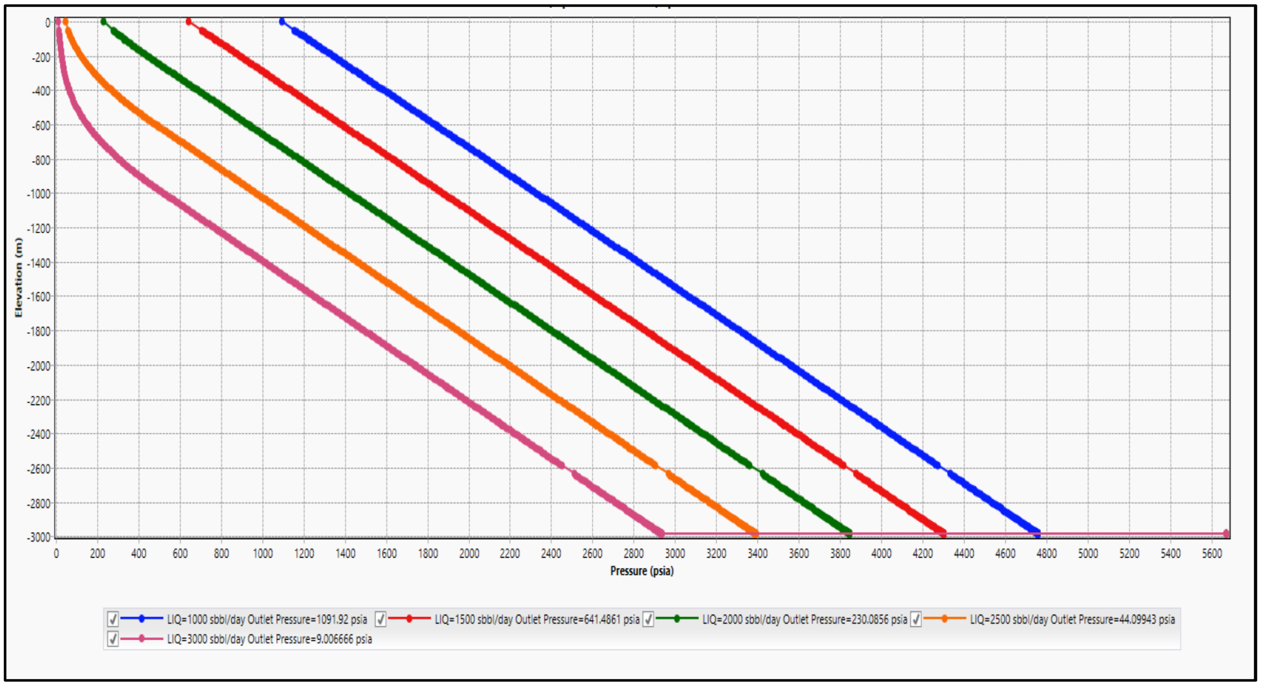


Figure 26—WHP vs. rate plot after selective acidizing.

From an engineering standpoint, these Multi phase flow software results confirm that selective acidizing successfully reduced skin and restored partial near-wellbore connectivity, setting the stage for the subsequent matrix acidizing phase. The selective stage achieved its design intent: clearing localized plugging and preparing the wellbore for deeper acid invasion, while maintaining operational stability and mechanical integrity.

After-acid-2 well test modeling

Following the full matrix acidizing stage, a final Multi phase flow software model was built to capture the ultimate productivity of well JR-09 under fully restored near-wellbore conditions. Actual flowing survey data collected after the treatment was incorporated into the model, and an iterative history-matching process was performed to ensure that simulated bottomhole pressures, pressure gradients, and flow profiles accurately matched the field measurements. Among several trial calibrations, the model configuration that provided the closest alignment was selected.

This stage marked the most significant change in the well's performance, with the Productivity Index (PI) rising to 2.24 STB/d/psi — more than double the PI after selective acidizing and over twice the pre-acid baseline value.

The final Inflow Performance Relationship (IPR) curve (Fig. 27) indicates that the back-pressure IPR nearly overlaps with the theoretical Darcy line, confirming that the matrix acidizing operation effectively eliminated almost all formation damage.

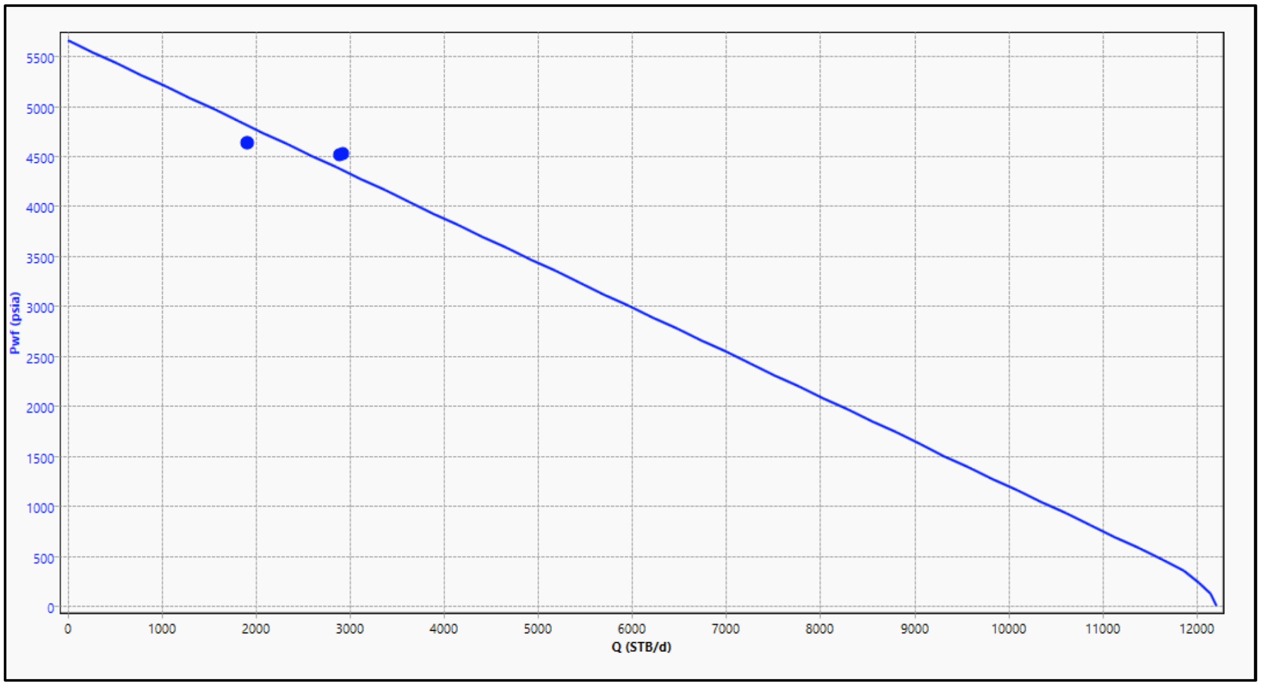


Figure 27—IPR curve after matrix acidizing showing improved inflow capacity.

A detailed history-matching exercise between the Multi phase flow software back-pressure IPR and the theoretical Darcy curve (Fig. 28) confirmed that the post-matrix skin factor reached −4.5. This negative skin value signifies not only the full removal of prior formation damage but also an enhanced flow condition around the wellbore. A negative skin implies an effectively enlarged wellbore radius and reduced flow convergence losses, which directly explains the substantial productivity gains and higher PI values observed after the treatment.

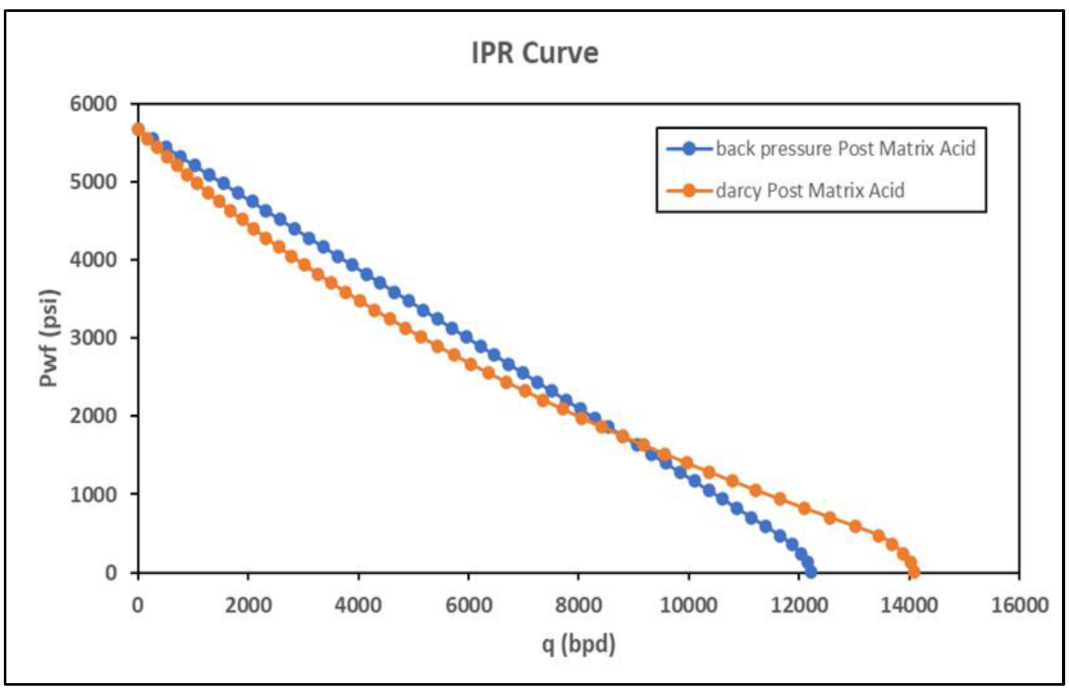


Figure 28—History match between back-pressure IPR and Darcy curve after matrix acidizing, confirming skin factor of −4.5 and showing stimulated wellbore performance.

The pressure profile (Fig. 29) shows an even, linear pressure decline from reservoir to wellhead with no steep drops, confirming that the frictional and near-wellbore pressure losses seen in the earlier models have largely been removed. The temperature profile (Fig. 30) remains smooth and progressive, declining from 208 °F in the reservoir to around 155–160 °F at the wellhead, with no thermal instability noted, even at the higher post-matrix flow rates.

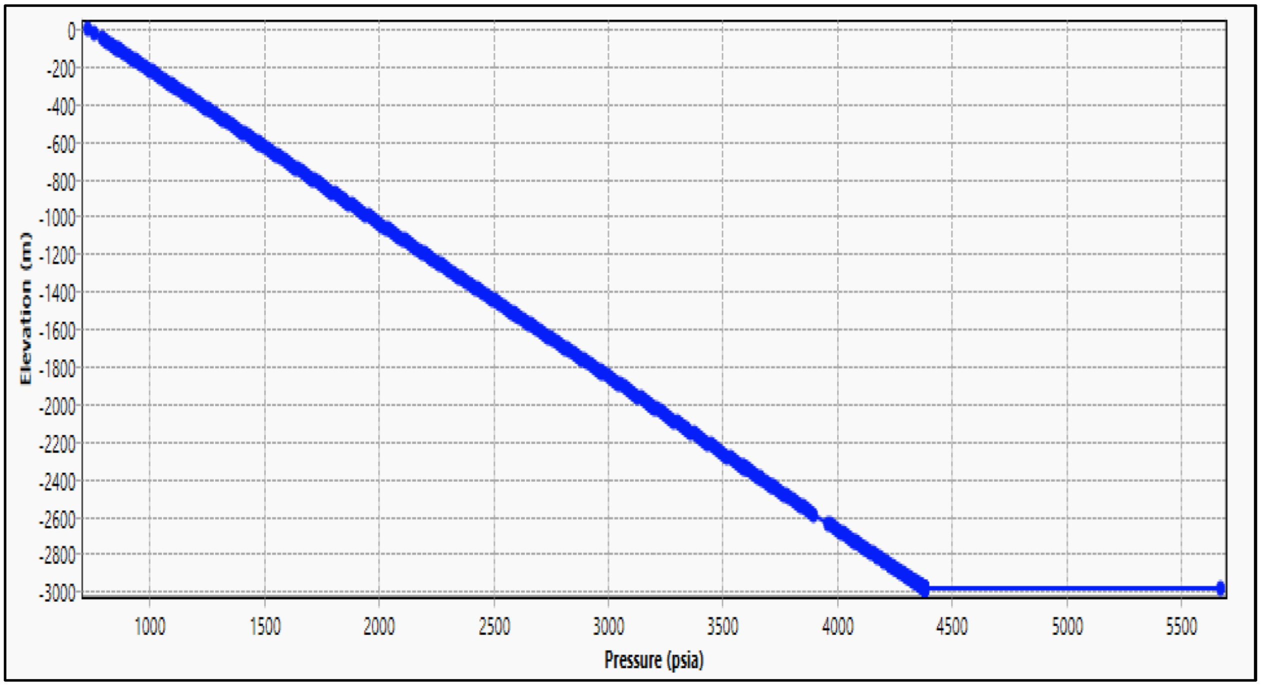


Figure 29—Pressure profile after matrix acidizing showing smooth pressure gradient and minimal near-wellbore losses.

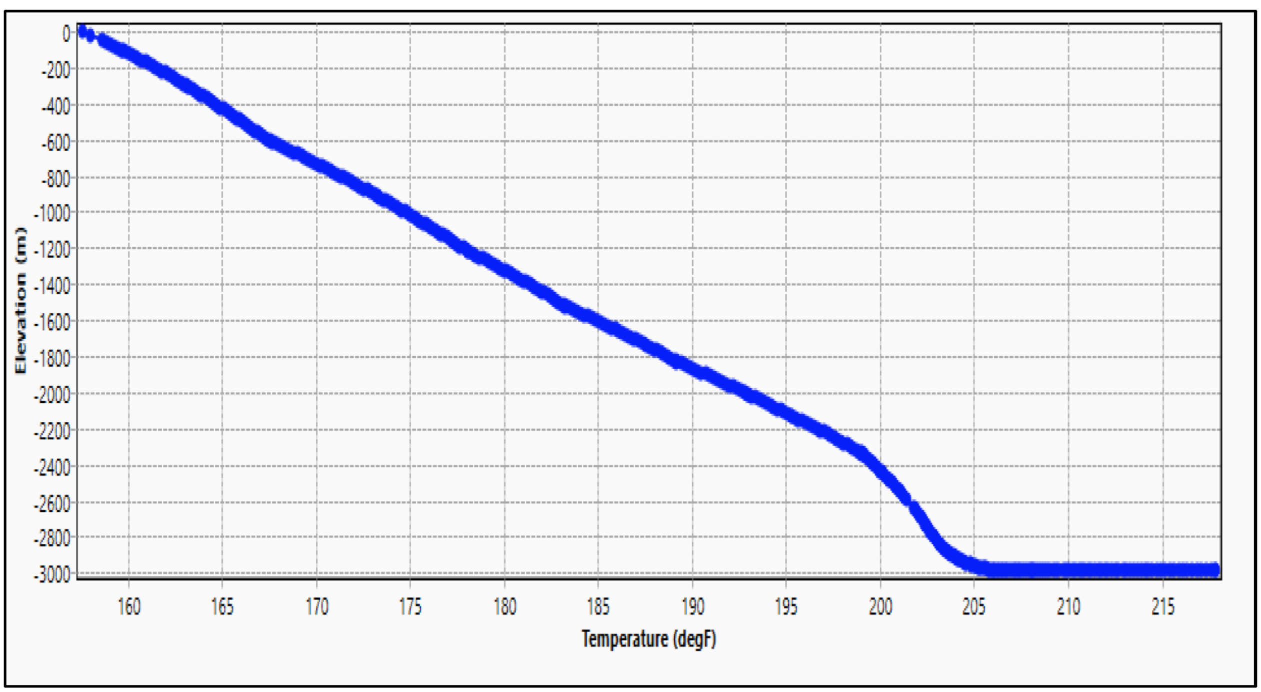


Figure 30—Temperature profile after matrix acidizing illustrating stable thermal behavior.

The WHP vs. rate data (Table 4 and Fig. 31) further validates the well's improved capacity. Unlike the pre-acid case, where WHP collapsed to near-atmospheric levels at modest rates, the post-matrix model shows positive wellhead pressure sustained even at 5,000 STB/d. This demonstrates that the matrix acidizing stage unlocked the full potential of JR-09 by providing deep wormhole penetration and improved reservoir connectivity.

Table 4—Post-Matrix Acidizing WHP vs. Rate

Rate (STB/d)	Wellhead Pressure (psia)
2000	1120.5
3000	681.3
4000	277.2
5000	65.0

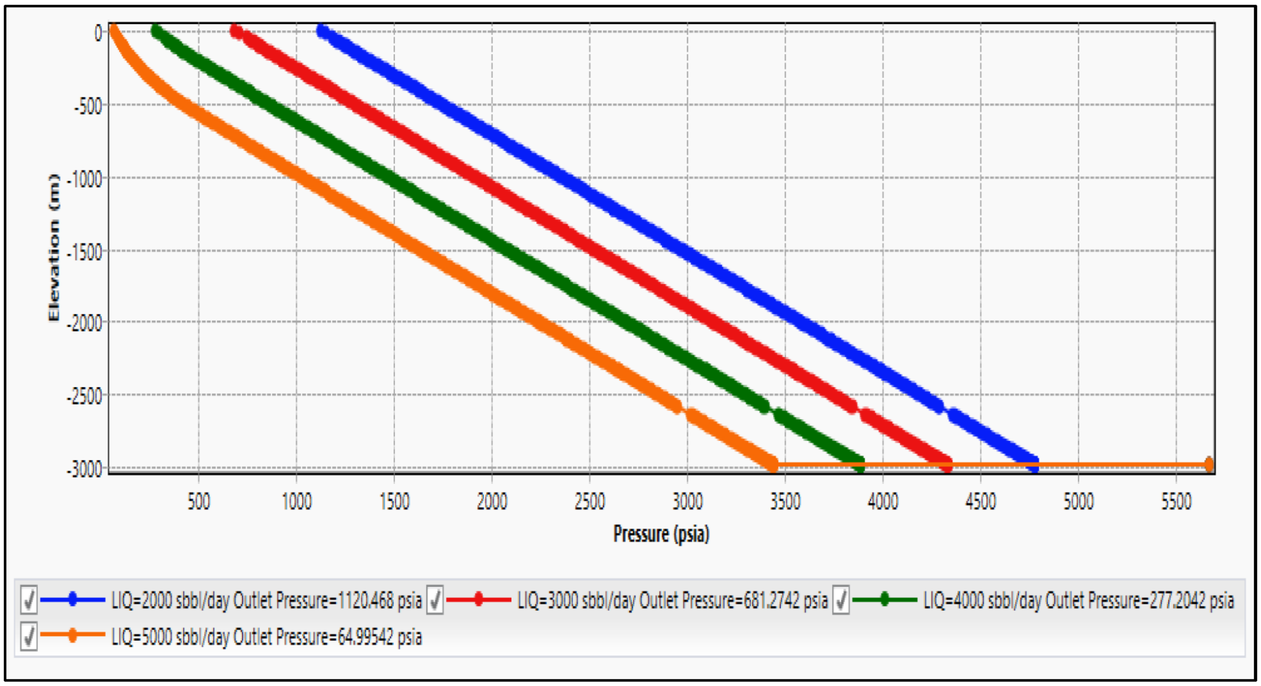


Figure 31—WHP vs. rate plot after matrix acidizing confirming sustained wellhead pressure at high flow rates.

From a reservoir engineering perspective, this stage completed the stimulation strategy: the selective phase had prepared the wellbore by removing local blockages, and the matrix phase extended stimulation deep into the reservoir, resulting in long-term productivity gains and near-ideal well performance.

The performance comparison of well JR-09 before and after the dual-stage acidizing operation provides a clear picture of the treatment's effectiveness. By evaluating the inflow performance relationship (IPR) curves from three stages — pre-acid, post-selective acidizing, and post-matrix acidizing — the progressive improvement in deliverability and well productivity becomes evident. This analysis highlights the impact of each stimulation phase and validates the combined mechanical-chemical diversion strategy applied in JR-09.

The IPR comparison (Fig. 32) shows a dramatic enhancement in well inflow capacity across the three phases. The pre-acid curve exhibits a steep drawdown requirement, with a calculated skin factor of +4.5 and a low productivity index (PI) of just 0.95 STB/d/psi. Following the selective acidizing phase, the skin factor decreased to approximately +3.0 and PI improved to 1.1 STB/d/psi — a clear indication that localized near-wellbore damage had been partially removed, although the deeper reservoir still restricted flow. The most significant transformation occurred after the matrix acidizing stage. The IPR curve shifted further outward, reflecting a PI increase to 2.24 STB/d/psi and a negative skin factor of −4.5. A negative skin signifies an effectively enlarged wellbore radius and confirms that matrix acidizing not only removed all remaining formation damage but also created a stimulated condition around the wellbore. This progressive change across the three IPR curves underscores the value of the dual-stage design: selective acidizing restored localized inflow capacity, and matrix acidizing extended stimulation deep into the reservoir, unlocking the full deliverability potential of JR-09.

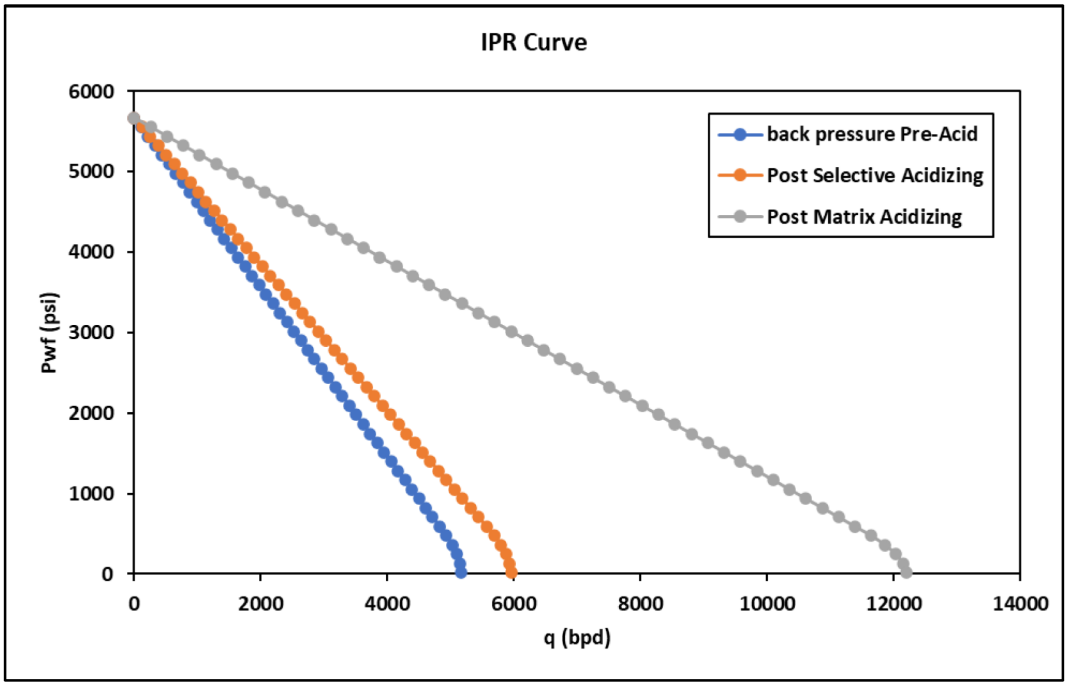


Figure 32—Comparison of IPR curves for JR-09 before stimulation, after selective acidizing, and after matrix acidizing.

From a petroleume engineering perspective, the comparison demonstrates that combining mechanical diversion via inflatable packers with chemical diversion and deep matrix acidizing is an effective strategy for treating complex dual-leg horizontal wells. The significant gains in PI and the transition from a damaged state (positive skin) to a fully stimulated condition (negative skin) provide a strong technical justification for applying this workflow in similar carbonate reservoirs.

Dual Leg Modeling. A comprehensive dual-leg Multi phase flow software network model was developed to evaluate the performance of well X-1 and quantify the impact of each stimulation stage on both laterals (Fig. 33). The model replicated the complete wellbore geometry, including the 4.5-inch vertical tubing to 2,349 m, the 7-inch liner down to the liner shoe at 3,073 m, and a tie-in junction at 3,225 m where flow splits into two horizontal branches. Each lateral — Leg 1 and Leg 2 — was represented as an independent reservoir node with distinct reservoir pressure, temperature, productivity index (PI), and skin values. This configuration enabled Multi phase flow software to calculate pressure, temperature, and flow contributions from each lateral separately before merging flows at the junction and delivering them to the wellhead.

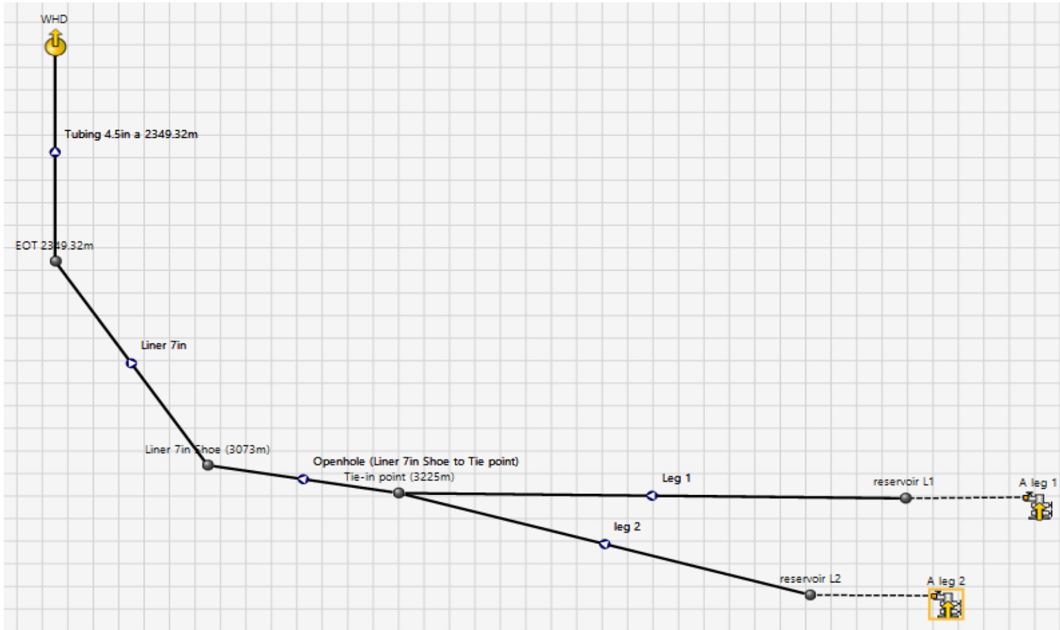


Figure 33—Multi phase flow software dual-leg network model of X-1 showing tubing, liner, tie-in junction, and two independent reservoir branches (Leg 1 and Leg 2).

Prior to stimulation, the model was calibrated using measured well test data. The baseline case matched a wellhead pressure of 185 psia and a total liquid production rate of 1,792 STB/d. Both laterals were assigned a reservoir pressure of 5,670 psia and temperature of 208 °F, with PI values of 9.0×10^{-5} STB/d/psi/ft for Leg 1 and 4.8×10^{-5} STB/d/psi/ft for Leg 2. The simulation confirmed a clear productivity imbalance: Leg 1 dominated inflow while Leg 2 contributed only marginally (Table 5).

Table 5—Pre-acid Multi phase flow software model output showing lower productivity from Leg 2 compared to Leg 1 and baseline WHP of 185 psia.

Name	Type	Pressure (psia)	Temperature (°F)	ST Liquid Rate (STB/d)	ST Oil Rate (STB/d)	ST Water Rate (STB/d)	ST Gas Rate (mmscfd)	ST GOR (SCF/STB)
WHD	Sink	185	157.8992	1792.682	1792.682	0	0.02602856	230
A Leg 1	Well	5670	208	1367.363	1367.363	0	0.3144935	230
A leg 2	Well	5670	208	419.4394	419.4394	0	0.09647107	230

Following the selective acidizing stage, the model was updated to reflect improved near-wellbore conditions. The wellhead pressure increased to 495 psia, while the total liquid production remained essentially unchanged (1,786 STB/d) as this stage mainly focused on debris removal and access preparation. PI values for Leg 1 (9.0×10^{-5}) and Leg 2 (4.8×10^{-5} STB/d/psi/ft) were unchanged, consistent with the operation's purpose of preparing the laterals for the matrix acidizing phase (Table 6).

Table 6—Post-selective acid Multi phase flow software model showing improved productivity (higher WHP) but no significant change in lateral PI values.

Name	Type	Pressure (psia)	Temperature (°F)	ST Liquid Rate (STB/d)	ST Oil Rate (STB/d)	ST Water Rate (STB/d)	ST Gas Rate (mmscfd)	ST GOR (SCF/STB)
WHD	Sink	494.7	140.35	1786.68	1786.68	0	0.410936	230
A Leg 1	Well	5670	208	1366.991	1366.991	0	0.3144079	230
A leg 2	Well	5670	208	419.326	419.326	0	0.09644498	230

The final update incorporated matrix acidizing results, delivering a step-change in well performance. The wellhead pressure increased further to 883 psia, and total liquid production surged to 2,930 STB/d. Both laterals showed substantial PI gains: Leg 1 rose by 63% (to 1.47×10^{-4} STB/d/psi/ft) and Leg 2 by 62% (to 7.8×10^{-5} STB/d/psi/ft), reflecting effective wormhole propagation and enhanced reservoir connectivity across the A carbonate formation (Table 7).

Table 7—Post-matrix Multi phase flow software model results showing significant PI uplift in both laterals and a production increase to 2,930 STB/d.

Name	Type	Pressure (psia)	Temperature (°F)	ST Liquid Rate (STB/d)	ST Oil Rate (STB/d)	ST Water Rate (STB/d)	ST Gas Rate (mmscfd)	ST GOR (SCF/STB)
WHD	Sink	882.7	144.3371	2929.774	2929.774	0	0.673848	230
A Leg 1	Well	5670	208	2244.654	2244.654	0	0.5162704	230
A leg 2	Well	5670	208	685.1199	685.1199	0	0.1575776	230

The dual-leg Multi phase flow software modeling workflow provided critical insights at every stage — from diagnosing pre-acid productivity imbalances, to validating the selective acidizing cleanup effect, to quantifying the major productivity uplift from matrix acidizing. The staged modeling confirmed that the dual-phase stimulation program successfully balanced leg contributions, increased total production, and delivered a measurable improvement in well performance.

Acid Stimulation Software Modeling. Comprehensive Acid Stimulation Software modeling was performed for all three acidizing phases of well JR-09 – the selective acidizing treatments on Leg 1 and Leg 2, and the full matrix acidizing stage – using actual field injection rates, surface pressures, and stage design volumes. This data-driven approach enabled the model to be history-matched to field conditions, ensuring that simulated pressure responses, fluid placement, and skin reduction predictions reflected actual well behavior rather than theoretical assumptions.

For Leg 2 selective acidizing, the model simulated approximately 196 bbl of 15% HCl, bracketing it with preflush and overflush stages as per the operational program. The simulation predicted a skin reduction from 5.0 to 2.4, with over 60% of this improvement occurring during the main acid stage (Fig. 34). The pressure match between measured and calculated values (Fig. 35) demonstrated excellent alignment, confirming that the model correctly captured rate-dependent friction losses in the coiled tubing, pressure drop across the inflatable packer, and the hydraulics of staged pumping. The invasion profile (Fig. 36) revealed penetration depths of 25–30 inches, concentrated around the isolated interval, confirming that mechanical isolation and controlled injection delivered acid precisely where intended, without bypassing into undesired zones.

For Leg 1 selective acidizing, a slightly larger volume of approximately 390 bbl of 15% HCl was modeled, again with preflush and overflush bracketing the acid stage. The simulation predicted skin reduction from 5.0 to 3.5 (Fig. 37). The skin-time curve indicated diminishing returns beyond the second main acid stage – an important finding suggesting that additional HCl beyond the designed volume would not have yielded proportional benefits. The pressure history match (Fig. 38) again demonstrated excellent correlation, validating that Acid Stimulation Software successfully modeled both frictional effects and the dynamic load on the inflatable packer system. The invasion profile (Fig. 39) indicated penetration up to 30 inches, demonstrating effective near-wellbore cleanup, but also highlighting that stimulation remained localized – exactly as designed for a selective phase – and did not over-etch or create uncontrolled channels.

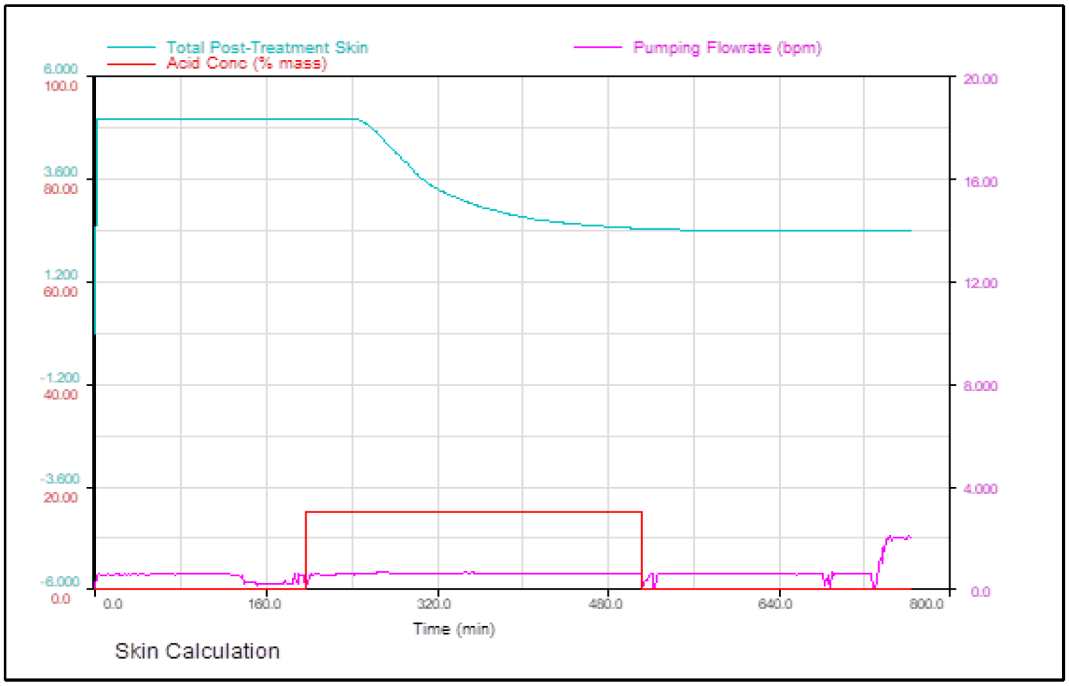


Figure 34—Skin calculation for Leg 2 selective acidizing showing skin reduction from 5.0 to 2.4.

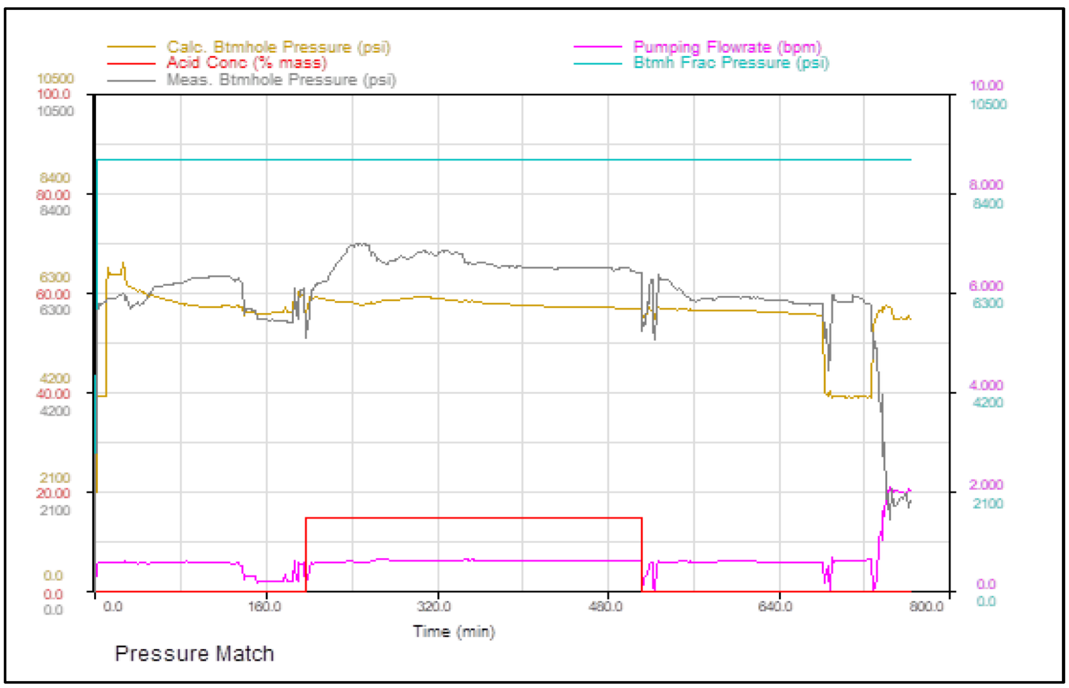


Figure 35—Pressure match for Leg 2 selective acidizing demonstrating alignment between simulated and measured data.

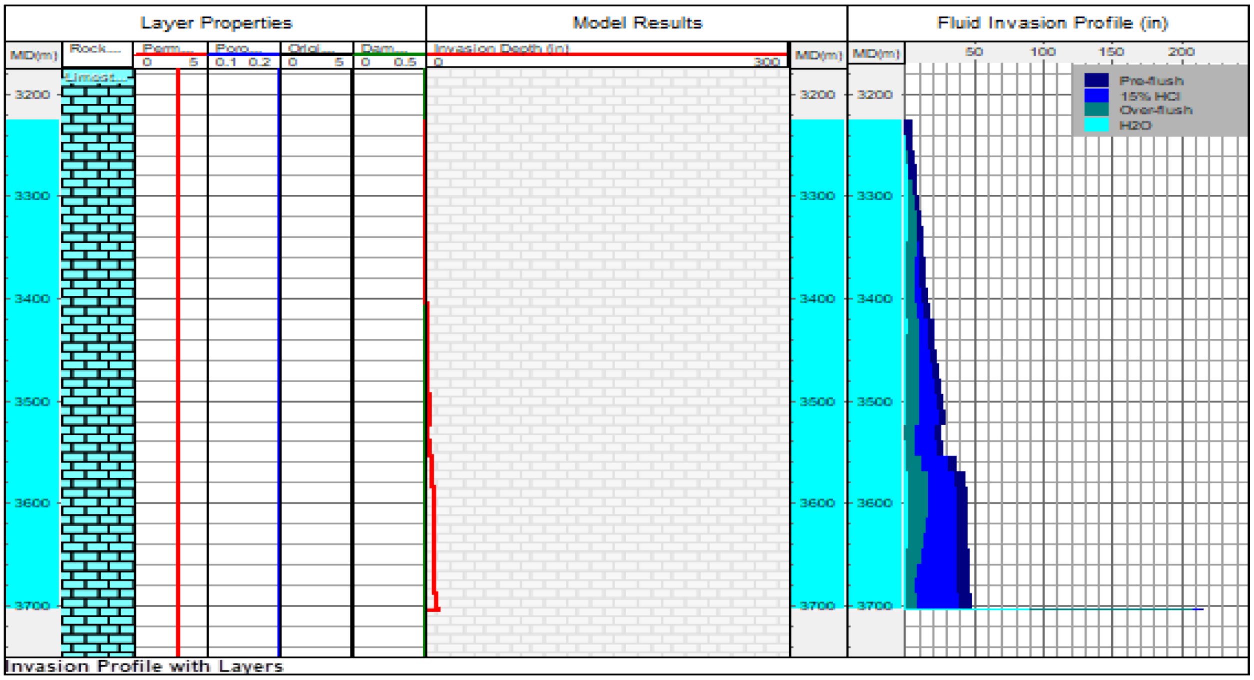


Figure 36—Fluid invasion profile for Leg 2 showing penetration of 25–30 inches in targeted zones.

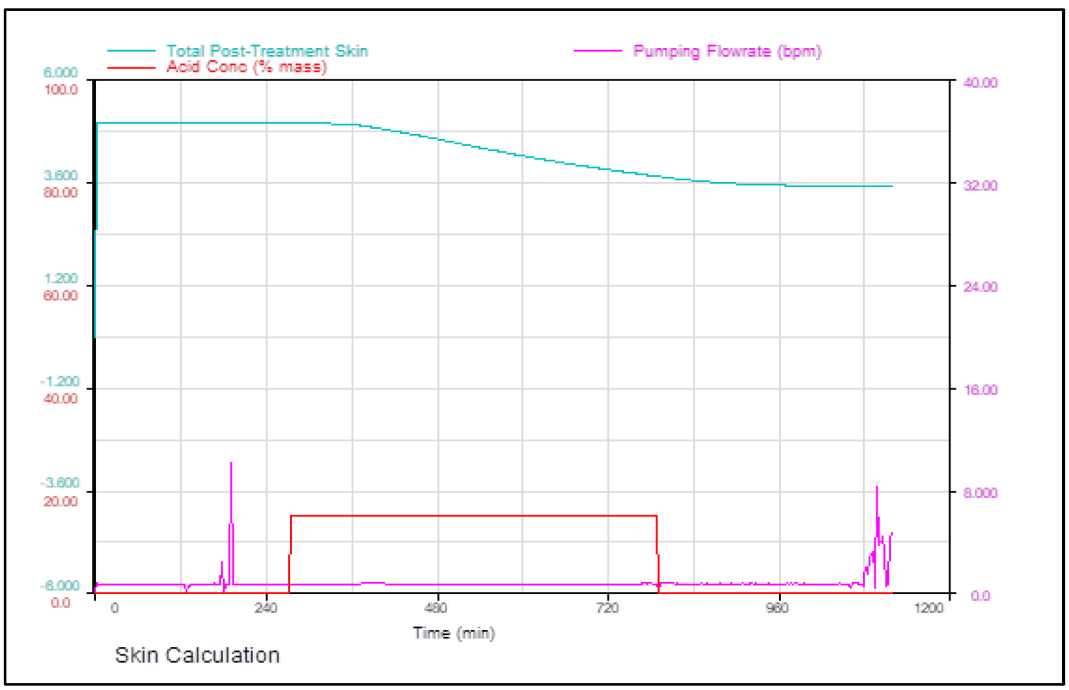


Figure 37—Skin calculation for Leg 1 selective acidizing showing skin reduction from 5.0 to 3.5.

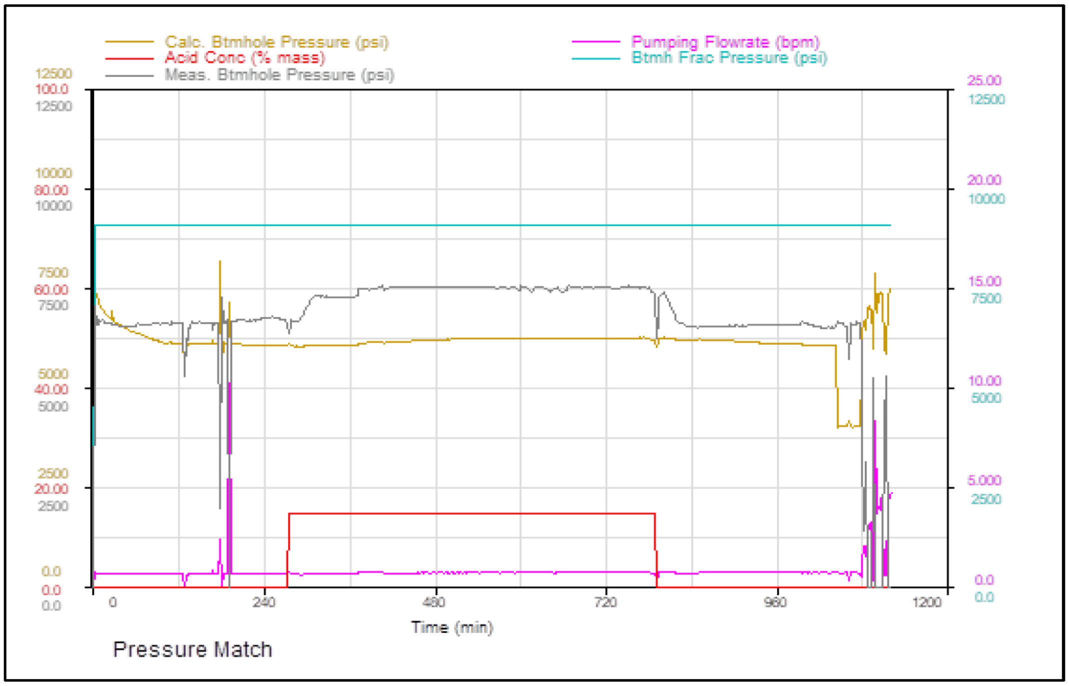


Figure 38—Pressure match for Leg 1 selective acidizing validating the model's calibration.

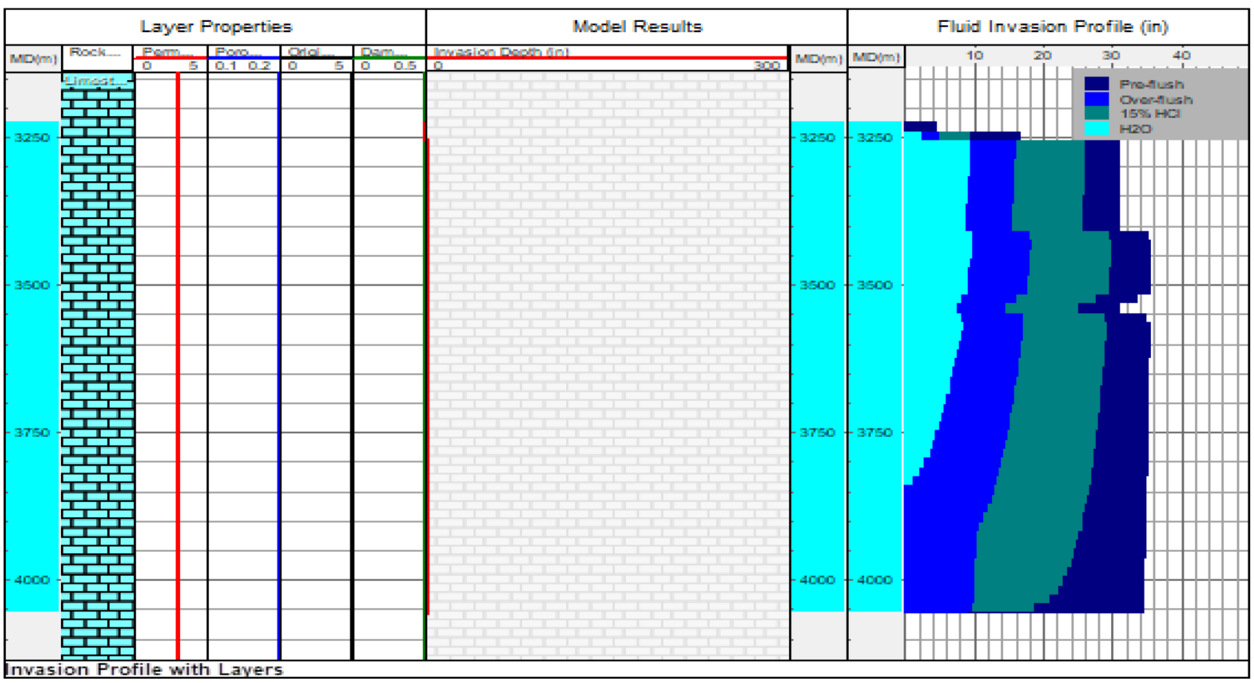


Figure 39—Fluid invasion profile for Leg 1 showing approximately 30 inches of acid penetration.

The matrix acidizing phase – by far the largest treatment – was modeled using the full nine-stage design: preflush, alternating 15% HCl and viscoelastic diverting acid (VDA) stages, and final overflush. Approximately 1,926 bbl of fluids were modeled (including 1,172 bbl HCl and 596 bbl VDA). The model predicted complete damage removal, with the skin index dropping from 2.50 to −1. 78 (Fig. 40). The pressure match (Fig. 41) again demonstrated that modeled friction losses, diverter effects, and pressure transients were faithfully reproduced – critical for verifying that the model could be trusted for design extrapolation. The invasion profile (Fig. 42) showed wormhole propagation exceeding 500 inches in several layers, confirming that the HCl/VDA alternating sequence efficiently diverted flow from already etched pathways into untreated zones, ensuring deep and uniform stimulation coverage.

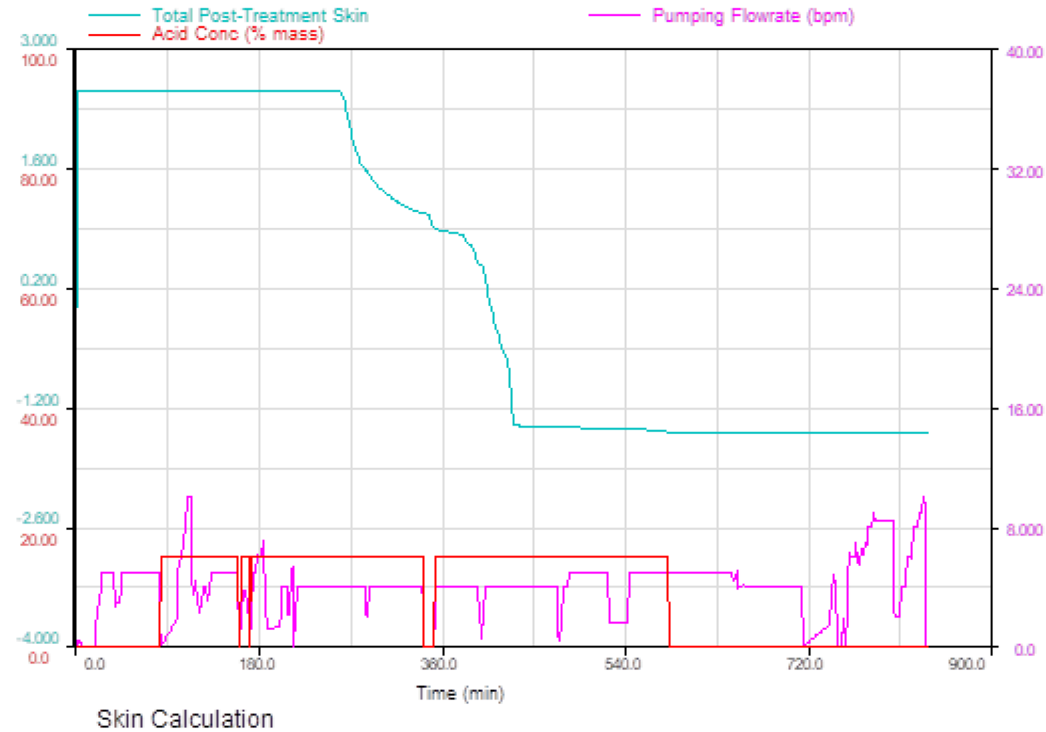


Figure 40—Skin calculation for matrix acidizing showing complete removal of damage (skin from 2.5 to −1.78).

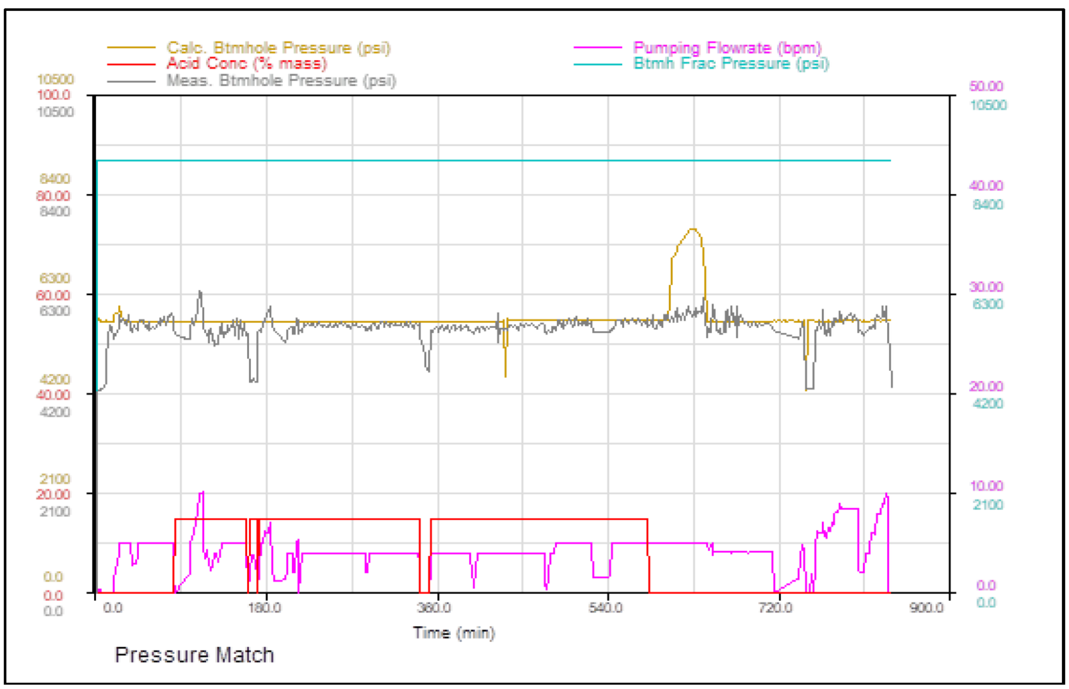


Figure 41—Pressure match for matrix acidizing showing accurate history matching with field data.

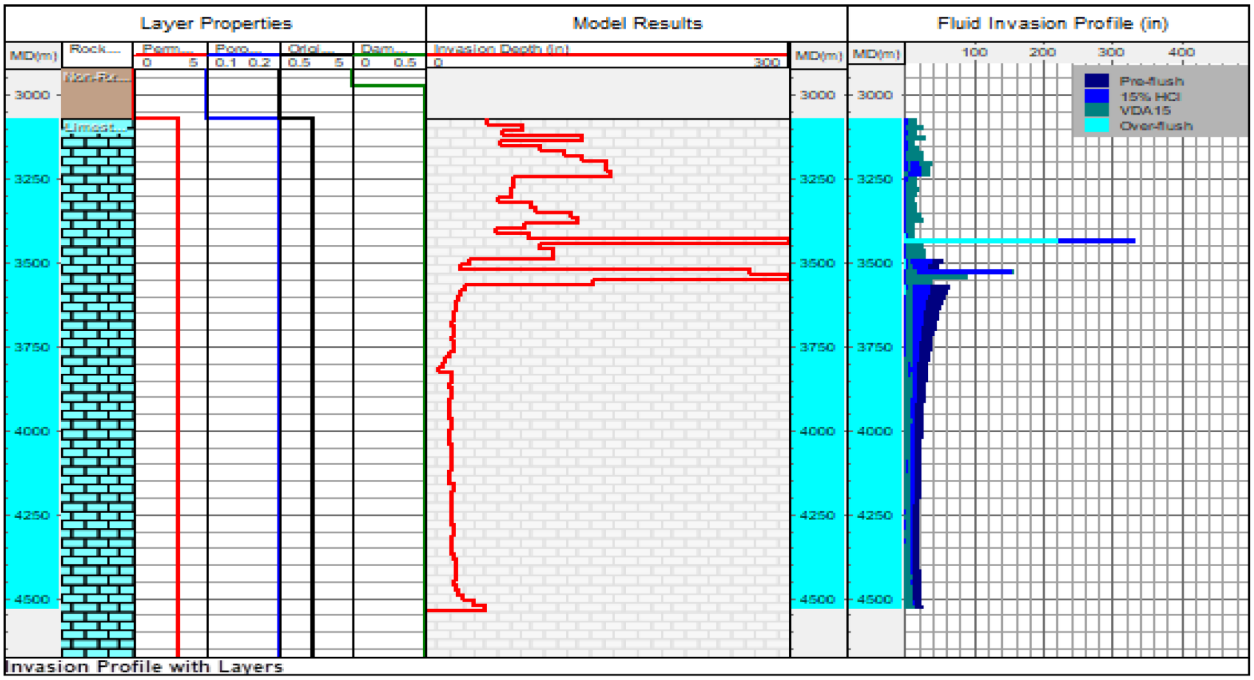


Figure 42—Fluid invasion profile for matrix acidizing revealing wormhole propagation beyond 500 inches.

From an engineering perspective, these simulations served multiple purposes. They validated that mechanical isolation with the inflatable packer in selective phases achieved precisely targeted stimulation and avoided wasteful acid diversion into non-targeted rock. They quantified the marginal gains of each stage, providing insights into how future treatments might optimize volumes for cost-effectiveness. They confirmed that the matrix phase achieved reservoir-wide cleanup, extending stimulation far beyond the near-wellbore region and fully restoring permeability.

Overall, the calibrated Acid Stimulation Software models bridge the gap between design and execution. They demonstrate that the dual-leg stimulation strategy – selective near-wellbore cleanup via CT and packer isolation, followed by deep matrix acidizing with chemical diversion – delivered the intended outcomes. These models now serve as a validated design framework for future dual-leg carbonate acidizing campaigns, providing both confidence in predicted performance and a reference point for refining fluid schedules, stage volumes, and diversion strategies.

Surface Well Testing Comparison. The performance comparison of well X-1 before and after acid stimulation clearly demonstrates the effectiveness of the dual-stage stimulation design. Multi-rate injectivity testing and surface well testing results confirm substantial improvements in both injectivity and productivity following the selective and matrix acidizing phases.

Before stimulation, the injectivity index (II) averaged only 2.1 bbl/d/psi, reaching a maximum of 2.54 bbl/d/psi at 5 bpm. After the selective acidizing stage, performed via coiled tubing with inflatable packers, the II nearly tripled to 6.99 bbl/d/psi, with notable improvements at injection rates of 2–5 bpm. This confirms that localized near-wellbore damage in both laterals was successfully removed, improving initial reservoir contact.

The matrix acidizing stage produced an even greater increase, with the II peaking at 11.71 bbl/d/psi at 10 bpm and maintaining an average of 9.24 bbl/d/psi across all test rates. These gains reflect deeper acid penetration and increased effective permeability, particularly in the low-permeability zones of the A carbonate reservoir. The injected volumes—approximately 1,271 bbl during selective acidizing and 2,637 bbl during matrix acidizing—were designed based on Acid Stimulation Software simulations to maximize coverage and wormhole propagation, which is reflected in the post-job injectivity trend (Fig. 43).

Before Acid Wash thru TBG				
Rate	Surface Pumping Pressure	average DHIP	ΔP = DHIP - P_R	Average Injectivity Index
bbl/m	psi	psi	psi	bpd/psi
1	1500	6,080	930	1.55
2	2100	6,680	1,530	1.88
3	2500	7,080	1,930	2.24
4	2850	7,430	2,280	2.53
5	3400	7,980	2,830	2.54
After Acid Wash with CT				
Rate	Surface Pumping Pressure	average DHIP	ΔP = DHIP - P_R	Average Injectivity Index
bbl/m	psi	psi	psi	bpd/psi
1	1150	5,730	580	2.48
2	1180	5,760	610	4.72
3	1255	5,835	685	6.31
4	1400	5,980	830	6.94
5	1600	6,180	1,030	6.99
After Matrix Acid thru TBG				
Rate	Surface Pumping Pressure	average DHIP	ΔP = DHIP - P_R	Average Injectivity Index
bbl/m	psi	psi	psi	bpd/psi
2	1210	5,790	640	4.50
4	1300	5,880	730	7.89
6	1395	5,975	825	10.48
8	1555	6,135	985	11.70
10	1800	6,380	1,230	11.71

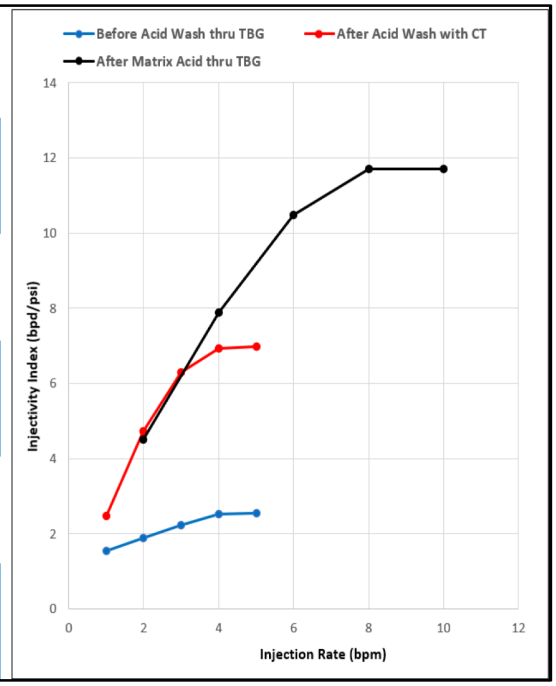


Figure 43—Injectivity Index (II) before and after stimulation stages.

Surface productivity results reinforce these findings. Pre-stimulation, the productivity index (PI) measured 1.75 bbl/d/psi, with an oil rate of 1,796 bbl/d at a wellhead pressure (WHP) of 171 psi on a 36/64-inch choke. After selective acidizing, PI increased to 2.44 bbl/d/psi, while production remained close to 1,750 bbl/d due to the smaller 28/64-inch choke size. The most significant improvement occurred after matrix acidizing: PI surged to 9.41 bbl/d/psi, and production exceeded 2,900 bbl/d even on tighter chokes (24–32/64-inch), demonstrating a marked enhancement in formation deliverability (Fig. 44).

Before Acid Wash		
Rate	WHP	Surface PI
bpd	psi	bpd/psi
1796	171	1.75
After Acid Wash		
Rate	WHP	Surface PI
bpd	psi	bpd/psi
1740	371	2.10
1754	480	2.44
After Matrix Acid		
Rate	WHP	Surface PI
bpd	psi	bpd/psi
1920	996	9.41
1905	989	9.03
2923	882	9.19
2894	868	8.72

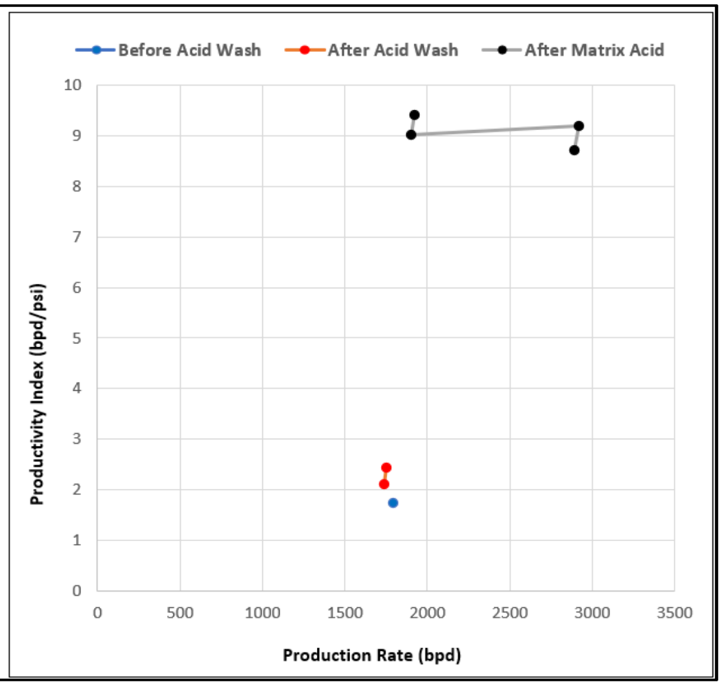


Figure 44—Surface Productivity Index (PI) vs. oil production rate for different stages.

The WHP–rate relationship further validates treatment effectiveness. Prior to stimulation, the well delivered 1,796 bbl/d with only 171 psi of WHP. After matrix acidizing, the well supported up to 2,894 bbl/d at WHP levels ranging from 868 to 996 psi (Fig. 45). The simultaneous increase in both WHP and production rate indicates substantial skin reduction and improved permeability across both laterals.

Before Acid Wash		
Rate	WHP	Choke Size
bpd	psi	in
1796	171	36.00

After Acid Wash		
Rate	WHP	Choke Size
bpd	psi	in
1740	371	32.00
1754	480	28.00

After Matrix Acid		
Rate	WHP	Choke Size
bpd	psi	in
1920	996	24.00
1905	989	24.00
2923	882	32.00
2894	868	32.00

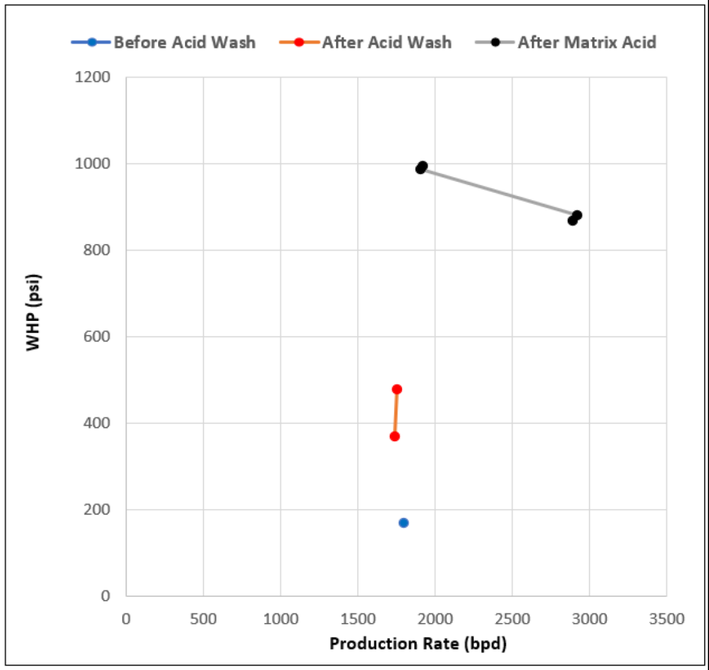


Figure 45—Wellhead Pressure (WHP) trend vs. oil production rate before and after stimulation.

Collectively, the injectivity, productivity, and WHP trends confirm a clear progression: selective acidizing removed near-wellbore blockages, while matrix acidizing established deep conductive channels. The combination of mechanical diversion (inflatable packers) and chemical diversion (VDA) ensured efficient fluid placement and full lateral coverage. This dual-leg stimulation strategy represents a scalable methodology for enhancing well performance in tight carbonate reservoirs.

These integrated results conclusively demonstrate that the dual-step selective and matrix acidizing approach — combining mechanical isolation via inflatable packers and chemical diversion using VDA — not only resolved near-wellbore damage but also created deep conductive channels, setting a new benchmark for stimulation strategies in complex dual-leg carbonate wells.

Conclusion

The dual-phase stimulation of well X-1 in the A carbonate reservoir has been shown to successfully overcome the inherent challenges of dual-leg open-hole horizontal wells through the integrated application of mechanical isolation and chemical diversion. The deployment of coiled tubing-conveyed inflatable packers facilitated precise zonal isolation during the selective acidizing phase, allowing targeted treatment of each lateral and effectively removing near-wellbore damage. This was followed by a matrix acidizing stage using 15% hydrochloric acid (HCl) and viscoelastic diverting acid (VDA), which delivered deep acid penetration and confirmed—through laboratory and field validation—the ability of VDA to achieve efficient diversion across heterogeneous carbonate intervals.

Extensive modeling and risk management underpinned the success of this project. Acid Stimulation Software simulations confirmed wormhole propagation and optimized injection parameters, while Coiled Tubing Hydraulic and Stress Analysis Simulator analysis validated mechanical integrity of the coiled tubing string under all operational loads. A dual-leg Multi phase flow software model quantified inflow improvements, showing that modeled productivity index (PI) increased from 0.95 STB/D/psi pre-stimulation to 2.24 STB/D/psi post-matrix acidizing, closely matching the surface PI increase from 1.75 to 8.7 STB/D/psi. The modeling also verified that the skin factor decreased from +5 to −4.5, confirming the removal of formation damage and restoration of natural reservoir inflow capacity.

The field operation was executed under a strict Management of Change (MOC) process, ensuring that all risk scenarios—including packer malfunction, CT lock-up, or acid misplacement—were anticipated and mitigated. As a result, the entire campaign was completed without any safety incidents, mechanical failures, or unplanned downtime. Additionally, the precise acid placement enabled by inflatable packers significantly reduced chemical volumes compared to a bullhead-only treatment, improving operational efficiency and reducing overall cost and environmental footprint.

Post-stimulation performance confirmed the technical and economic success of the treatment. The injectivity index (II) increased fivefold, wellhead pressure rose by over 700 psi, and oil production climbed from 1,800 BOPD to approximately 2,900 BOPD. The verified diversion performance of VDA ensured uniform treatment coverage, and nitrogen lift support during clean-up operations facilitated effective well unloading, preventing any post-treatment damage.

This project marks the first documented field application in Iran of a dual-leg selective and matrix acidizing approach using open-hole inflatable packers. The methodology—anchored in simulation-driven design, laboratory validation, disciplined execution, and rigorous risk management—establishes a benchmark for future stimulation campaigns in the A reservoir and similar tight carbonate systems worldwide. It demonstrates that combining mechanical isolation for selective stages with chemical diversion for matrix phases can deliver superior productivity gains, reduced chemical consumption, and improved operational safety, offering a scalable, cost-effective solution for complex horizontal carbonate wells.

References

1. H. A. Nasr-El-Din and H. Y. Al-Yousef, "A Novel Technique to Acidize Multilateral Open Hole ER Horizontal Wells Drilled in Carbonate Formations," in SPE Middle East Oil and Gas Show and Conference, 2005.
2. G. Glasbergen and M. Buijse, "Improved Acid Diversion Design Using a Placement Simulator," in SPE Annual Technical Conference and Exhibition, 2006.
3. L. Jin, "Optimising Diversion and Pumping Rate To Effectively Stimulate Long Horizontal Carbonate Gas Wells," in SPE European Formation Damage Conference, 2007.
4. G. Glasbergen, V. Yeager, R. Reyes, and D. Everett, "Fluid-Diversion Monitoring: The Key to Treatment Optimization," *SPE Production & Operations*, vol. **25**, no. 03, pp. 262–274, Aug. 2010, doi: 10.2118/122353-PA.
5. A. Sarma and coauthors, "Innovative Stimulation Approach in a Mature Field with Special Reference to Multilayered, Horizontal and Multilateral Wells," *SPE Production & Operations*, 2012.
6. H. A. Nasr-El-Din, J. B. Chesson, K. E. Cawiezel, and C. S. Devine, "Lessons Learned and Guidelines for Matrix Acidizing With Viscoelastic Surfactant Diversion in Carbonate Formations," Proceedings - SPE Annual Technical Conference and Exhibition, vol. **3**, pp. 2003–2013, Sep. 2006, doi: 10.2118/102468-MS.
7. M. A. Al-Otaibi, G. A. Al-Muntasheri, I. A. Hussein, and F. F. Chang, "Experimental Evaluation of Viscoelastic Surfactant Acid Diversion for Carbonate Reservoirs: Parameters and Performance Analysis," SPE Middle East Oil and Gas Show and Conference, MEOS, Proceedings, vol. **3**, pp. 1530–1546, Sep. 2011, doi: 10.2118/141993-MS.

8. F. F. Chang, A. M. Acock, A. Geoghagan, and P. T. Huckabee, "Experience in Acid Diversion in High Permeability Deep Water Formations Using Visco-Elastic-Surfactant," SPE - European Formation Damage Control Conference, Proceedings, pp. 77–81, May 2001, doi: 10.2118/68919-MS.
9. G. Czupski and coauthors, "Development of Selective Acidizing Technology for an Oil Field in the Zechstein Main Dolomite," *MDPI Energies*, 2020.
10. G. Czupski and coauthors, "A Comprehensive Analysis of Carbonate Matrix Acidizing Using Viscoelastic Diverting Acid System in a Gas Field," *MDPI Energies*, 2024.
11. A. Alameedy and coauthors, "Well Performance Following Matrix Acidizing Treatment; Case Study of the Mi4 Unit in Ahdeb Oil Field," *Iraqi Journal of Chemical and Petroleum Engineering*, 2022.
12. M. Jeffry and coauthors, "Selection of Suitable Acid Chemicals for Matrix Stimulation; A Malaysian Brown Field Scenario," *Petroleum*, 2020.
13. O. G. Chacon and M. Pournik, "Matrix Acidizing in Carbonate Formations," *Processes 2022*, Vol. **10**, Page 174, vol. 10, no. 1, p. 174, Jan. 2022, doi: 10.3390/PR10010174.
14. M. Al Mahasneh and coauthors, "Assessment of Oil-Producing Wells by Means of Stimulation Approach Through Matrix Acidizing: A Case Study," *Geofluids*, 2021.
15. S. Al-Harthy, "*Options for High Temperature Well Stimulation*," 2008.
16. M. Buijse and G. Glasbergen, "A semiempirical model to calculate wormhole growth in carbonate acidizing," SPE Annual Technical Conference Proceedings, 2005, doi: 10.2118/96892-ms.
17. D. B. Bennion and F. B. Thomas, "Procedures for Minimizing Drilling and Completion Damage in Horizontal Wells," *Journal of Canadian Petroleum Technology*, 1994.
18. R. Poyyara and coauthors, "Optimization of Acid Treatments by Assessing Diversion Strategies in Carbonate and Sandstone Formations," *International Journal of Chemical, Material and Natural Resources Engineering*, 2014.